dirac16complex: the Dirac equation and Kohn–Sham theory of a 16-component fermion field on 7+1 spacetime

Real 16×16 Dirac matrices, the chiral 8-component equations, and the 16-component and chiral Kohn–Sham theories of the self-coupled field — sections 7–10 of the notebook 7+1.nb

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Abstract

This document defines and verifies (i) the Dirac equation of a 16-component quantized fermion field named dirac16complex on the real 7+1 spacetime (one time coordinate x_0, seven space coordinates $x_1, \ldots, x_7$), whose components are complex-valued anticommuting (Grassmann) operators, with eight anticommuting REAL 16×16 Dirac matrices; (ii) the chiral projections that split the 16-component equation into two 8-component equations, coupled by the mass; (iii) a 16-component Kohn–Sham theory of the field coupled to itself through scalar, pseudoscalar, vector and axial-vector contact interactions; (iv) the chiral (8+8-component) form of that Kohn–Sham theory; (v) worked examples: uniform matter in the Hartree and Hartree–Fock approximations, with saturation and the Hugenholtz–Van Hove theorem, and the chiral vacuum with the Dirac sea (spontaneous chiral symmetry breaking). Sections 7–9 of 7+1.nb contain 74 checks and the whole notebook 479; all are True, and an independent Python program reproduces every identity and every number.

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1. Results at a glance

Real Dirac matrices. With the real 8×8 octonion matrices τ_i^O ($i = 1, \dots, 7$; antisymmetric, $(\tau_i^O)^2 = -1_8$, mutually anticommuting; section 2.3) the eight Dirac matrices are

$$\Gamma_0 = \begin{pmatrix} 0 & 1_8 \\ 1_8 & 0 \end{pmatrix}, \qquad \Gamma_i = \begin{pmatrix} \tau_i^O & 0 \\ 0 & -\tau_i^O \end{pmatrix}, \qquad \Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\,\eta_{AB}\,1_{16}, \qquad \eta = \mathrm{diag}(+1, -1, \dots, -1).$$

Every entry is 0 or ±1. The signs are those of the octonion basis: η is simply the squares of the octonion units, $e_0^2 = +1$ (time) and $e_i^2 = -1$ (space), $\eta_{AB} = \mathrm{Re}(e_A e_B)$ (section 2.1). These are the only real choice: with the opposite sign convention (time squaring to -1) no real 16×16 matrices exist (section 3.3).

The Dirac equation of dirac16complex $\psi = (\psi_1, \dots, \psi_{16})$, with $\Gamma^A = \eta^{AA}\Gamma_A$ and $\partial_A = \partial/\partial x_A$:

$$\big(i\,\Gamma^A\partial_A - m\big)\psi = 0 \qquad \Longleftrightarrow \qquad i\,\partial_0\psi = \big(-i\,\alpha_i\,\partial_i + \beta\,m\big)\psi, \qquad \beta = \Gamma^0, \quad \alpha_i = \Gamma^0\Gamma^i = \begin{pmatrix} 0 & \tau_i^O \\ -\tau_i^O & 0 \end{pmatrix}.$$

Two coupled 8-component equations. There is no real chirality matrix in 7+1 dimensions; the chirality $\chi = i\,\Gamma_0\Gamma_1\cdots\Gamma_7$ is complex, and so are the chiral projections $P_{L,R} = \frac{1}{2}(1_{16} \mp \chi)$. With $\phi_L = (\psi_\uparrow + i\psi_\downarrow)/\sqrt{2}$ and $\phi_R = (i\psi_\uparrow + \psi_\downarrow)/\sqrt{2}$ ($\psi_\uparrow, \psi_\downarrow$ the upper and lower 8 components):

$$\big(i\,\partial_0 + \tau_i^O\,\partial_i\big)\,\phi_L = m\,\phi_R,$$
$$\big(i\,\partial_0 - \tau_i^O\,\partial_i\big)\,\phi_R = m\,\phi_L.$$

Only the real τ_i^O appear; the mass couples the two 8-component fields, which decouple for $m = 0$.

16-component Kohn–Sham equations of the field coupled to itself (couplings g_S, g_P, g_V, g_A):

$$h_{\mathrm{KS}}\,\varphi_k = \varepsilon_k\,\varphi_k,$$
$$h_{\mathrm{KS}} = \alpha_i\big(-i\partial_i + \mathcal{V}_i\big) + \beta\big(m + \mathcal{S}\big) + \mathcal{V}_0 + \beta\,i\chi\,\mathcal{P} + \chi\,\mathcal{A}_0 + \alpha_i\chi\,\mathcal{A}_i,$$
$$\mathcal{S} = \mathsf{s} + g_S\rho_S + \frac{\delta E_{xc}}{\delta\rho_S}, \qquad \mathcal{P} = \mathsf{p} + g_P\rho_P + \frac{\delta E_{xc}}{\delta\rho_P},$$

$$\mathcal{V}_A = \mathtt{v}_A + g_V j_A + \frac{\delta E_{xc}}{\delta j^A}, \qquad \mathcal{A}_A = \mathtt{a}_A + g_A j_{5A} + \frac{\delta E_{xc}}{\delta j_5^A}.$$

Chiral Kohn–Sham equations (8+8 components), with the complex Kohn–Sham mass $\mathcal{M} = m + \mathcal{S} + i\mathcal{P}$ and $\mathcal{V}_A^{L,R} = \mathcal{V}_A \mp \mathcal{A}_A$, $\sigma_i = i\tau_i^O$:

$$\big[\mathcal{V}_0^L - \sigma_i\big(-i\partial_i + \mathcal{V}_i^L\big)\big]\varphi_L + \mathcal{M}\,\varphi_R = \varepsilon\,\varphi_L,$$
$$\mathcal{M}^*\,\varphi_L + \big[\mathcal{V}_0^R + \sigma_i\big(-i\partial_i + \mathcal{V}_i^R\big)\big]\varphi_R = \varepsilon\,\varphi_R.$$

Examples. Uniform matter with $g_S = -10^5$, $g_V = 10^4$ (units $m = 1$, no Dirac sea) saturates at $k_F = 0.756636150702$ with $E/A - m = -0.0283214279647$ and $m^* = 0.348097340146$; the Hugenholtz–Van Hove theorem $E/A = \varepsilon_F$ holds there to 25 digits. With the Dirac sea, cutoff $\Lambda = 1$ and $g_S = g_P = \frac{6}{5}g_c$, $g_c = -90\pi^4/\Lambda^6$, chiral symmetry breaks spontaneously: $|\mathcal{M}| = 0.553693851829$.

Verification. 74 checks in sections 7–9 of 7+1.nb (479 in the whole notebook), all True, no message cells; independently, tools/ref_dirac16.py (Python, sympy/mpmath/numpy) passes 90 checks and reproduces all printed numbers to 17 digits (section 11).

2. Spacetime, units, notation

2.1 Coordinates and metric

Units $\hbar = c = 1$. The coordinates are the eight real numbers $X = (x_0, x_1, \ldots, x_7)$ (Mathematica: `X = {x0, x1, x2, x3, x4, x5, x6, x7}`): x_0 is time (the direction of the octonion unit $e_0 = 1$), x_1, x_2, x_3 are ordinary 3-space and $x_4, \ldots, x_7$ are four auxiliary space-like dimensions. The metric is

$$\eta_{AB} = \eta^{AB} = \mathrm{diag}(+1, -1, -1, -1, -1, -1, -1, -1), \qquad a \cdot b = \eta_{AB}\, a^A b^B = a^0 b^0 - \sum_{i=1}^{7} a^i b^i,$$

the real part of the octonion square, $\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2$. Indices $A, B, K, L \in \{0, \ldots, 7\}$ are raised and lowered with η; $i, j \in \{1, \ldots, 7\}$ are space indices; repeated indices are summed. $\partial_A = \partial/\partial x_A$ (the coordinates are $x^A = x_A$ as symbols, and ∂_A acts on them as $\partial_A x^B = \delta_A^B$). A space vector is written with an arrow, $\vec{p} = (p^1, \ldots, p^7)$, $\vec{p}\cdot\vec{x} = \sum_i p^i x_i$, $\vec{p}^{\,2} = \sum_i (p^i)^2$. The wave (d'Alembert) operator is

$$\Box = \eta^{AB}\partial_A\partial_B = \partial_0^2 - \partial_1^2 - \cdots - \partial_7^2.$$

The signature: $\{7, 1, 0\}$ and $\mathrm{diag}(+1, -1, \ldots, -1)$. The 7+1 spacetime has one time-like dimension (x_0, the direction of the octonion unit $e_0 = 1$) and seven space-like dimensions (x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space). Two different objects describe this, and neither may be replaced by the other. (i) The tensor package xAct, used in section 12 of 7+1.nb exactly as in the author's notebook Einstein-Lovelock-7+1-Nash.nb, declares the metric with `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`: the list $\{7, 1, 0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension — the numbers of entries $+1$ and -1 of $\eta' = \mathrm{diag}(-1, +1, \ldots, +1)$ (`η71`), the same spacetime written in the sign convention of 4+4.nb (space-like = $+1$, time-like = -1). (ii) The matrix $\eta = -\eta' = \mathrm{diag}(+1, -1, \ldots, -1)$ (`η17`) is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units $e_1, \ldots, e_7$ (space); more generally it is the real part of the product of two basis units, and it gives the quadratic form of the octonions:

$$\eta_{AB} = \mathrm{Re}(e_A e_B), \qquad e_0^2 = +1, \qquad e_i^2 = -1 \;\; (i = 1, \ldots, 7),$$
$$\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = \eta_{AB}\, x^A x^B \qquad (x = x^A e_A \in \mathbb{O}).$$

η gives the signs of the Clifford relation $\Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\,\eta_{AB}\,1_{16}$ of the real 16×16 Dirac matrices: time squares to $+1_{16}$ and each of the seven space directions to -1_{16}, exactly like the octonion units (in the notation $Cl(p, q)$, with p generators squaring to $+1$ and q to -1, the algebra $Cl(1, 7) \cong \mathbb{R}(16)$). It also gives the signs of the metric components: $g_{00} > 0$ along the time-like direction, negative squares along the space-like directions. Every component computed by 7+1.nb follows from the component matrix of the metric; the declared list fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` $= -1$, which is the same in both sign conventions, since $\det\eta = \det\eta' = -1$. 7+1.nb checks `SignatureOfMetric[g71]` $= \{7, 1, 0\}$, `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$ and $\mathrm{Re}(x\,x) = \eta_{AB} x^A x^B$.

2.2 Matrix notation

1_n is the $n\times n$ identity, M^{T} the transpose, $M^\dagger = \overline{M}^{\mathsf{T}}$ the Hermitian conjugate, $\overline{z}$ or z^* the complex conjugate, tr the trace, δ_{ij} the Kronecker delta and $\{X,Y\} = XY + YX$. A 16×16 matrix is often written as 2×2 blocks of 8×8 matrices, $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, and a 16-component column as $(\psi_\uparrow;\psi_\downarrow)$ (upper and lower 8 components). e_s $(s = 1,\dots,8)$ is the s-th unit column of length 8. Mathematica names are given in `typewriter`; Mathematica lists are 1-based, so η_{AB} is `η17[[A+1, B+1]]` in the notebook.

2.3 The real 8×8 matrices τ_i^O

The building blocks are the seven real 8×8 matrices $\tau_1^O,\dots,\tau_7^O$ of 7+1.nb (`τ0[i]`; $\tau_0^O = 1_8$). They are the left multiplications by the imaginary octonion units $e_1,\dots,e_7$ written in a spinor basis, and they satisfy

$$(\tau_i^O)^{\mathsf{T}} = -\tau_i^O, \qquad \tau_i^O\tau_j^O + \tau_j^O\tau_i^O = -2\,\delta_{ij}\,1_8, \qquad \tau_1^O\tau_2^O\cdots\tau_7^O = 1_8 \qquad (i,j = 1,\dots,7).$$

$$\tau_1^O = \begin{pmatrix} 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&1&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&0&0&-1&0&0&0\\ 0&0&0&1&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0\\ -1&0&0&0&0&0&0&0 \end{pmatrix}, \quad \tau_2^O = \begin{pmatrix} 0&0&0&0&0&0&-1&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&-1&0&0&0&0&0\\ 0&0&0&1&0&0&0&0\\ 1&0&0&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0 \end{pmatrix},$$

$$\tau_3^O = \begin{pmatrix} 0&0&0&0&0&1&0&0\\ 0&0&0&0&-1&0&0&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&-1&0\\ 0&1&0&0&0&0&0&0\\ -1&0&0&0&0&0&0&0\\ 0&0&0&1&0&0&0&0\\ 0&0&-1&0&0&0&0&0 \end{pmatrix}, \quad \tau_4^O = \begin{pmatrix} 0&1&0&0&0&0&0&0\\ -1&0&0&0&0&0&0&0\\ 0&0&0&-1&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&-1&0 \end{pmatrix},$$

$$\tau_5^O = \begin{pmatrix} 0&0&-1&0&0&0&0&0\\ 0&0&0&-1&0&0&0&0\\ 1&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ 0&0&0&0&0&0&1&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&-1&0&0&0\\ 0&0&0&0&0&-1&0&0 \end{pmatrix}, \quad \tau_6^O = \begin{pmatrix} 0&0&0&-1&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0\\ 1&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&-1&0\\ 0&0&0&0&0&1&0&0\\ 0&0&0&0&-1&0&0&0 \end{pmatrix},$$

$$\tau_7^O = \begin{pmatrix} 0&0&0&0&-1&0&0&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&0&0&0&0&-1&0\\ 0&0&0&0&0&0&0&-1\\ 1&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&0&0&1&0&0&0&0 \end{pmatrix}.$$

Where they come from. 7+1.nb builds them from the real split-octonion matrices τ_A of the notebook 4+4.nb: $\tau_i^O = \tau_i$ for $i = 1,2,3$ and $\tau_i^O = \tau_i\,\sigma$ for $i = 4,\dots,7$, where $\sigma = \tau_1\tau_2\tau_3 = \left(\begin{smallmatrix} 0 & 1_4 \\ 1_4 & 0 \end{smallmatrix}\right)$ commutes with τ_1,τ_2,τ_3, anticommutes with $\tau_4,\dots,\tau_7$ and squares to 1_8, so that $(\tau_k\sigma)^2 = -\tau_k^2\sigma^2 = -1_8$ for $k = 4,\dots,7$

(the split matrices $\tau_4,\dots,\tau_7$ square to $+1_8$). Explicitly, the four split matrices are

$$\tau_4 = \begin{pmatrix} 0&0&0&0&0&1&0&0\\ 0&0&0&0&-1&0&0&0\\ 0&0&0&0&0&0&0&-1\\ 0&0&0&0&0&0&1&0\\ 0&-1&0&0&0&0&0&0\\ 1&0&0&0&0&0&0&0\\ 0&0&0&1&0&0&0&0\\ 0&0&-1&0&0&0&0&0 \end{pmatrix}, \quad \tau_5 = \begin{pmatrix} 0&0&0&0&0&0&-1&0\\ 0&0&0&0&0&0&0&-1\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&1&0&0\\ 0&0&1&0&0&0&0&0\\ 0&0&0&1&0&0&0&0\\ -1&0&0&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0 \end{pmatrix},$$

$$\tau_6 = \begin{pmatrix} 0&0&0&0&0&0&0&-1\\ 0&0&0&0&0&0&1&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&0&0&1&0&0&0\\ 0&0&0&1&0&0&0&0\\ 0&0&-1&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ -1&0&0&0&0&0&0&0 \end{pmatrix}, \quad \tau_7 = \begin{pmatrix} -1&0&0&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0\\ 0&0&-1&0&0&0&0&0\\ 0&0&0&-1&0&0&0&0\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&1&0&0\\ 0&0&0&0&0&0&1&0\\ 0&0&0&0&0&0&0&1 \end{pmatrix},$$

and τ_1,τ_2,τ_3 equal $\tau_1^O,\tau_2^O,\tau_3^O$ displayed above.

With $\sigma_i = i\,\tau_i^O$ one obtains seven Hermitian 8×8 matrices with $\sigma_i\sigma_j+\sigma_j\sigma_i = 2\delta_{ij}1_8$: the analogues of the Pauli matrices for seven space dimensions (Mathematica `σW[i]`). For a space vector, $\tau^O\cdot\vec p = \sum_i p^i\tau_i^O$ and $\sigma\cdot\vec p = \sum_i p^i\sigma_i$, with $(\tau^O\cdot\vec p)^2 = -\vec p^{\,2}1_8$ and $(\sigma\cdot\vec p)^2 = \vec p^{\,2}1_8$.

3. The eight real Dirac matrices

3.1 Definition and Clifford relation

The Dirac matrices are the real 16×16 matrices Γ_A (`Γ17[A]`, section 4.2 of 7+1.nb):

$$\Gamma_0 = \begin{pmatrix} 0 & 1_8 \\ 1_8 & 0 \end{pmatrix}, \qquad \Gamma_i = \begin{pmatrix} \tau_i^O & 0 \\ 0 & (\tau_i^O)^{\mathsf T} \end{pmatrix} = \begin{pmatrix} \tau_i^O & 0 \\ 0 & -\tau_i^O \end{pmatrix} \quad (i=1,\dots,7),$$

$$\Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\,\eta_{AB}\,1_{16} \qquad (A,B=0,\dots,7): \qquad \Gamma_0^2 = +1_{16}, \quad \Gamma_i^2 = -1_{16}.$$

Every entry is 0 or ±1. In terms of the 16×16 Dirac matrices of 4+4.nb,

$$T_A = \begin{pmatrix} 0 & \bar\tau_A \\ \tau_A & 0 \end{pmatrix}, \qquad \bar\tau_0 = 1_8, \quad \bar\tau_A = \sigma\,\tau_A^{\mathsf T}\sigma \ (A=1,\dots,7),$$

$$T_AT_B + T_BT_A = 2\,\eta_{AB}^{(4,4)}\,1_{16}, \qquad \eta^{(4,4)} = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1),$$

(with τ_A and σ of section 2.3) they are $\Gamma_0 = T_0$, $\Gamma_i = T_0T_i$ $(i=1,2,3)$ and $\Gamma_k = T_0T_kT_4T_5T_6T_7$ $(k=4,\dots,7)$. The index is raised with η:

$$\Gamma^A = \eta^{AA}\,\Gamma_A: \qquad \Gamma^0 = \Gamma_0, \qquad \Gamma^i = -\Gamma_i, \qquad \Gamma^A\Gamma_A = 8\cdot 1_{16}.$$

Because Γ_0 is symmetric, the Γ_i are antisymmetric and all are real,

$$\left(\Gamma^A\right)^\dagger = \left(\Gamma^A\right)^{\mathsf T} = \Gamma^0\,\Gamma^A\,\Gamma^0.$$

The 256 ordered products $\Gamma_K = \Gamma_{A_1}\Gamma_{A_2}\cdots\Gamma_{A_k}$ $(K=\{A_1<\cdots<A_k\}\subseteq\{0,\dots,7\},\ \Gamma_\emptyset = 1_{16})$ are linearly independent, so Γ_A generate the full real Clifford algebra $Cl(1,7)\cong\mathbb{R}(16)$ of the quadratic form $Q(v)=v\cdot v$: $\left(\sum_A v^A\Gamma_A\right)^2 = Q(v)\,1_{16}$.

3.2 The Dirac α, β and the volume element ω

$$\beta = \Gamma^0 = \begin{pmatrix} 0 & 1_8 \\ 1_8 & 0 \end{pmatrix}, \qquad \alpha_i = \Gamma^0\Gamma^i = -\Gamma_0\Gamma_i = \begin{pmatrix} 0 & \tau_i^O \\ -\tau_i^O & 0 \end{pmatrix} \quad (i = 1, \ldots, 7),$$

$$\alpha_i\alpha_j + \alpha_j\alpha_i = 2\,\delta_{ij}\,1_{16}, \qquad \alpha_i\beta + \beta\alpha_i = 0, \qquad \beta^2 = 1_{16}, \qquad \alpha_i^{\mathsf{T}} = \alpha_i, \quad \beta^{\mathsf{T}} = \beta \ \text{(real symmetric)},$$

$$\omega = \Gamma_0\Gamma_1\cdots\Gamma_7 = \begin{pmatrix} 0 & -1_8 \\ 1_8 & 0 \end{pmatrix}, \qquad \omega^2 = -1_{16}, \qquad \omega\,\Gamma_A = -\Gamma_A\,\omega \quad (A = 0, \ldots, 7).$$

In the notebook: $\beta =$ `βD`, $\alpha_i =$ `αD[i]`, $\omega =$ `ω17` $=$ `Γ17[8]`.

3.3 Why real Dirac matrices must square like the octonion basis, $\eta = \mathrm{diag}(+1, -1, \ldots, -1)$

The opposite overall sign, $\eta' = -\eta = \mathrm{diag}(-1, +1, \ldots, +1)$ (time squaring to -1), generates the Clifford algebra $Cl(7,1) \cong \mathbb{H}(8)$, which has NO real 16×16 realization: if real R_A with $R_AR_B + R_BR_A = 2\eta'_{AB}1_{16}$ existed, then, $Cl_8(\mathbb{C}) \cong \mathbb{C}(16)$ having a single irreducible module, $R_A = T\,(i\Gamma_A)\,T^{-1}$ for an invertible complex T; reality of R_A gives $Y\,\overline{(i\Gamma_A)} = (i\Gamma_A)\,Y$ with $Y = T^{-1}\overline{T}$ and $Y\overline{Y} = 1_{16}$. Because the Γ_A are real, $\overline{(i\Gamma_A)} = -i\Gamma_A$, so $Y\Gamma_A = -\Gamma_A Y$ for all A; then $Y\omega$ commutes with every Γ_A and, by Schur's lemma (the commutant of the irreducible $\mathbb{C}(16)$ is $\mathbb{C}\,1_{16}$), $Y = c\,\omega$ with $c \in \mathbb{C}$ (the real solutions, computed in 7+1.nb, are $\mathbb{R}\,\omega$). But then $Y\overline{Y} = |c|^2\omega^2 = -|c|^2\,1_{16} \neq 1_{16}$ — a contradiction. Reordering, renaming or re-signing indices does not change the signature of the Clifford relation (the numbers of generators squaring to $+1$ and to -1), so it cannot help. Hence real 16×16 Dirac matrices of 7+1 spacetime must square like the octonion basis: $\eta = \mathrm{diag}(+1, -1, \ldots, -1)$ is simply the squares $e_0^2 = +1$, $e_i^2 = -1$ of the octonion units, and the Dirac matrices of this document are real. (The xAct list $\{7, 1, 0\}$ of section 12 of 7+1.nb is a different object: it declares seven space-like dimensions and one time-like dimension of the same spacetime.)

4. The field dirac16complex and its Dirac equation

4.1 The field

The quantized fermion field is the column of 16 components (Mathematica: `dirac16complex = Table[ψ16[a] @@ X, {a, 1, 16}]`)

$$\psi(x) \equiv \texttt{dirac16complex}(x) = \big(\psi_1(x), \ldots, \psi_{16}(x)\big)^{\mathsf{T}}, \qquad x = (x_0, x_1, \ldots, x_7).$$

Each component is a complex (non-Hermitian, $\psi_a^\dagger \neq \psi_a$) Grassmann-odd operator field. At equal times

$$\psi_a(x_0,\vec{x})\,\psi_b(x_0,\vec{y}) + \psi_b(x_0,\vec{y})\,\psi_a(x_0,\vec{x}) = 0, \qquad \psi_a(x_0,\vec{x})\,\psi_b^\dagger(x_0,\vec{y}) + \psi_b^\dagger(x_0,\vec{y})\,\psi_a(x_0,\vec{x}) = \delta_{ab}\,\delta^7(\vec{x}-\vec{y}),$$

in particular $\psi_a^2 = 0$ (section 6). The Dirac adjoint is $\bar{\psi} = \psi^\dagger\Gamma^0$.

4.2 The Dirac equation and the Klein–Gordon equation

$$\big(i\,\Gamma^A\partial_A - m\big)\psi = 0 \qquad \text{i.e.} \qquad i\,\Gamma_0\,\partial_0\psi - i\sum_{i=1}^{7}\Gamma_i\,\partial_i\psi - m\,\psi = 0.$$

The matrices are real; the factor i in front of them is unavoidable. With real Γ_A the first-order operator squares to $+\Box$: $(\Gamma^A\partial_A)^2 = \frac{1}{2}\{\Gamma^A,\Gamma^B\}\partial_A\partial_B = \eta^{AB}\partial_A\partial_B = \Box$, hence

$$\big(i\,\Gamma^A\partial_A + m\big)\big(i\,\Gamma^A\partial_A - m\big) = -\big(\Box + m^2\big)\,1_{16}\,:$$

every component of a solution obeys the 7+1 Klein–Gordon equation $(\Box + m^2)\psi_a = 0$. The equation is linear, so it reads the same for operator-valued, Grassmann-valued or ordinary complex components; 7+1.nb verifies the identities for 16 arbitrary functions $\psi_a(x_0, \ldots, x_7)$.

4.3 Hamiltonian form

Multiplying by Γ^0 from the left ($\Gamma^0\Gamma^0 = 1_{16}$):

$$i\,\partial_0\psi = h\,\psi, \qquad h = -i\,\alpha_i\,\partial_i + \beta\,m, \qquad \Gamma^0\big(i\Gamma^A\partial_A - m\big)\psi = \big(i\partial_0 - h\big)\psi.$$

α_i and β are real symmetric and $-i\partial_i$ is Hermitian, so h is Hermitian: the evolution is unitary and the charge $Q = \int \psi^\dagger\psi\, d^7x$ is conserved.

4.4 Lagrangian and conserved current

$$\mathcal{L}_0 = \bar\psi\big(i\,\Gamma^A\partial_A - m\big)\psi, \qquad j^A = \bar\psi\,\Gamma^A\psi, \qquad \partial_A j^A = 0.$$

Conservation uses the Dirac equation and its adjoint. The Hermitian conjugate of $(i\Gamma^A\partial_A - m)\psi = 0$ is $-i\partial_A\psi^\dagger(\Gamma^A)^\dagger - m\psi^\dagger = 0$; with $(\Gamma^A)^\dagger = \Gamma^0\Gamma^A\Gamma^0$ and $\psi^\dagger = \bar\psi\Gamma^0$ this is

$$-i\,\partial_A\bar\psi\,\Gamma^A - m\,\bar\psi = 0, \qquad \text{so} \qquad \partial_A\big(\bar\psi\Gamma^A\psi\big) = (\partial_A\bar\psi)\Gamma^A\psi + \bar\psi\Gamma^A\partial_A\psi = (im\bar\psi)\psi + \bar\psi(-im\psi) = 0.$$

The charge density is $j^0 = \psi^\dagger\psi$ and the spatial current $j^i = \psi^\dagger\alpha_i\psi$.

5. Plane waves and the real spinors u_s, v_s

5.1 The momentum-space Hamiltonian

For a plane wave $e^{-i(Ex_0 - \vec p\cdot\vec x)}$ the operator $-i\partial_i$ becomes p^i and h becomes the REAL symmetric 16×16 matrix

$$h(\vec p) = p^i\alpha_i + m\beta = \begin{pmatrix} 0 & M(\vec p) \\ M(\vec p)^{\mathsf T} & 0 \end{pmatrix}, \qquad M(\vec p) = m\,1_8 + p^i\tau_i^O \quad (\text{real } 8\times 8), \qquad M(-\vec p) = M(\vec p)^{\mathsf T}.$$

Because the τ_i^O are antisymmetric and anticommute, $(p^i\tau_i^O)(p^j\tau_j^O)^{\mathsf T} = -(p^i\tau_i^O)^2 = \vec p^{\,2}1_8$ and

$$MM^{\mathsf T} = M^{\mathsf T}M = E^2\,1_8, \qquad h(\vec p)^2 = E^2\,1_{16}, \qquad E = \sqrt{m^2 + \vec p^{\,2}}.$$

5.2 Real spinors

The eigenvalues of $h(\vec p)$ are $\pm E$, eight times each, with REAL eigenvectors: for $s = 1, \ldots, 8$

$$u_s(\vec p) = \frac{1}{\sqrt E}\begin{pmatrix} E\,e_s \\ M^{\mathsf T}e_s \end{pmatrix}, \qquad h(\vec p)\,u_s = +E\,u_s,$$

$$v_s(\vec p) = \frac{1}{\sqrt E}\begin{pmatrix} E\,e_s \\ -M\,e_s \end{pmatrix}, \qquad h(-\vec p)\,v_s = -E\,v_s,$$

$$u_s^{\mathsf T}u_r = v_s^{\mathsf T}v_r = 2E\,\delta_{sr}, \qquad u_s(\vec p)^{\mathsf T}v_r(-\vec p) = 0.$$

With the 8-momentum $p = (E, \vec p)$ and the real matrix $\not p = \Gamma_A p^A = E\,\Gamma_0 + p^i\Gamma_i$ these are the momentum-space Dirac equations

$$(\not p - m)\,u_s(\vec p) = 0, \qquad (\not p + m)\,v_s(\vec p) = 0,$$

and the plane waves $u_s(\vec p)\,e^{-i(Ex_0 - \vec p\cdot\vec x)}$ (positive energy) and $v_s(\vec p)\,e^{+i(Ex_0 - \vec p\cdot\vec x)}$ (negative energy) solve $(i\Gamma^A\partial_A - m)\psi = 0$. 7+1.nb proves all of this for a completely symbolic momentum $\vec p = (p_1, \ldots, p_7)$.

Example (printed by 7+1.nb): $\vec{p}=(2,2,4,0,0,0,0)$, $m=1$, $E=5$. The eight spinors u_s are the columns of

$$\sqrt{5}\,(u_1,\dots,u_8)=\begin{pmatrix}
5&0&0&0&0&0&0&0\\
0&5&0&0&0&0&0&0\\
0&0&5&0&0&0&0&0\\
0&0&0&5&0&0&0&0\\
0&0&0&0&5&0&0&0\\
0&0&0&0&0&5&0&0\\
0&0&0&0&0&0&5&0\\
0&0&0&0&0&0&0&5\\
1&0&0&0&0&-4&2&-2\\
0&1&0&0&4&0&-2&-2\\
0&0&1&0&-2&2&0&-4\\
0&0&0&1&2&2&4&0\\
0&-4&2&-2&1&0&0&0\\
4&0&-2&-2&0&1&0&0\\
-2&2&0&-4&0&0&1&0\\
2&2&4&0&0&0&0&1
\end{pmatrix},$$

and $h(\vec{p})\,u_s=5\,u_s$.

5.3 Spin sums, completeness and the diagonalization of h

Since the spinors are real, $\bar{u}_s=u_s^{\mathsf{T}}\Gamma^0$ and

$$\sum_{s=1}^{8}u_s\,\bar{u}_s=\not{p}+m,\qquad \sum_{s=1}^{8}v_s\,\bar{v}_s=\not{p}-m,$$

$$\sum_{s=1}^{8}u_s(\vec{p})\,u_s(\vec{p})^{\mathsf{T}}+\sum_{s=1}^{8}v_s(-\vec{p})\,v_s(-\vec{p})^{\mathsf{T}}=2E\,1_{16}.$$

Equivalently the real orthogonal matrix

$$U(\vec{p})=\frac{1}{\sqrt{2E}}\big(u_1(\vec{p}),\dots,u_8(\vec{p}),\,v_1(-\vec{p}),\dots,v_8(-\vec{p})\big),\qquad U^{\mathsf{T}}U=1_{16},\qquad U^{\mathsf{T}}h(\vec{p})\,U=\begin{pmatrix}E\,1_8&0\\0&-E\,1_8\end{pmatrix}$$

diagonalizes $h(\vec{p})$; this is what makes the mode expansion of section 6.2 work.

6. Grassmann variables, quantization, charge conjugation, Lorentz covariance

6.1 Grassmann algebra

Classically the components of dirac16complex are Grassmann numbers: generators $\theta_1,\theta_2,\dots$ with

$$\theta_k\theta_l=-\theta_l\theta_k\qquad(\text{so }\theta_k^2=0).$$

An element is a linear combination of ordered monomials $\theta_{k_1}\theta_{k_2}\cdots\theta_{k_r}$ ($k_1<\cdots<k_r$; in 7+1.nb `grass[{k1, ..., kr}]`, with `grass[{}]` $=1$ and $\theta_k=$ `θg[k]`). The product of two monomials is 0 if they share a generator, and otherwise the sorted monomial times the sign of the sorting permutation (`gMul`). The left derivative $\partial^L/\partial\theta_k$ (`gD`) moves θ_k to the front and removes it:

$$\frac{\partial^L}{\partial\theta_k}\big(\theta_{k_1}\cdots\theta_{k_r}\big)=(-1)^{j-1}\,\theta_{k_1}\cdots\widehat{\theta_{k_j}}\cdots\theta_{k_r}\quad\text{if }k=k_j,\qquad\text{e.g.}\quad\frac{\partial^L}{\partial\theta_2}\big(\theta_1\theta_2\theta_3\big)=-\theta_1\theta_3,$$

where the hat means omission. **Check of current conservation with a Grassmann-valued solution.** Take three plane-wave solutions with $m = 1$,

$$U_1 = u_1(\vec{p}_1)\, e^{-i(5x_0 - \vec{p}_1\cdot\vec{x})}, \quad U_2 = v_3(\vec{p}_3)\, e^{+i(3x_0 - \vec{p}_3\cdot\vec{x})}, \quad U_3 = u_5(\vec{0})\, e^{-ix_0},$$
$$\vec{p}_1 = (2,2,4,0,0,0,0), \qquad \vec{p}_3 = (0,0,0,2,2,0,0),$$

and the Grassmann-valued field and its adjoint

$$\psi = \theta_1 U_1 + \theta_2 U_2 + \theta_3 U_3, \qquad \bar{\psi} = \bar{\theta}_1\, U_1^\dagger \Gamma^0 + \bar{\theta}_2\, U_2^\dagger \Gamma^0 + \bar{\theta}_3\, U_3^\dagger \Gamma^0, \qquad \bar{\theta}_k = \theta_{3+k}.$$

7+1.nb verifies $\psi_a \psi_b + \psi_b \psi_a = 0$ for all a, b and $\partial_A(\bar{\psi}\Gamma^A \psi) = 0$ in this Grassmann algebra.

6.2 Mode expansion and canonical anticommutators

In a box of volume V with periodic boundary conditions the quantized field is

$$\psi(x_0, \vec{x}) = \sum_{\vec{p}} \sum_{s=1}^{8} \frac{1}{\sqrt{2E\,V}} \Big[b_s(\vec{p})\, u_s(\vec{p})\, e^{-i(Ex_0 - \vec{p}\cdot\vec{x})} + d_s^\dagger(\vec{p})\, v_s(\vec{p})\, e^{+i(Ex_0 - \vec{p}\cdot\vec{x})} \Big],$$

$$\big\{ b_s(\vec{p}), b_r^\dagger(\vec{q}) \big\} = \big\{ d_s(\vec{p}), d_r^\dagger(\vec{q}) \big\} = \delta_{sr}\, \delta_{\vec{p}\vec{q}}, \qquad \text{all other anticommutators of } b, b^\dagger, d, d^\dagger \text{ vanish.}$$

$b_s(\vec{p})$ annihilates a particle and $d_s^\dagger(\vec{p})$ creates an antiparticle, each with 8 spin states. The completeness relation of section 5.3 turns the mode algebra into the canonical anticommutators

$$\big\{ \psi_a(x_0, \vec{x}), \psi_b^\dagger(x_0, \vec{y}) \big\} = \delta_{ab}\, \frac{1}{V} \sum_{\vec{p}} e^{i\vec{p}\cdot(\vec{x} - \vec{y})} = \delta_{ab}\, \delta^7(\vec{x} - \vec{y}),$$
$$\big\{ \psi_a(x_0, \vec{x}), \psi_b(x_0, \vec{y}) \big\} = 0 \qquad (\psi \text{ contains only } b \text{ and } d^\dagger).$$

7+1.nb sums the mode algebra exactly, with symbolic $\vec{x}, \vec{y}$ and $m = 1$, for the symmetric set of five momenta

$$\vec{p} \in \big\{ \pm(2,2,4,0,0,0,0),\ \pm(0,0,0,2,2,0,0),\ \vec{0} \big\} \qquad (E = 5,5,3,3,1),$$

and obtains $\delta_{ab} \frac{1}{V} \sum_{\vec{p}} e^{i\vec{p}\cdot(\vec{x} - \vec{y})}$. Because $U^{\mathsf{T}} h U = \mathrm{diag}(E, -E)$, the normal-ordered Hamiltonian and charge are

$$H = \int : \psi^\dagger h\, \psi : \, d^7x = \sum_{\vec{p},s} E\, \big(b_s^\dagger b_s + d_s^\dagger d_s \big), \qquad Q = \int : \psi^\dagger \psi : \, d^7x = \sum_{\vec{p},s} \big(b_s^\dagger b_s - d_s^\dagger d_s \big) :$$

particles and antiparticles have the same positive energy $E = \sqrt{m^2 + \vec{p}^{\,2}}$ and opposite charge.

6.3 Charge conjugation

Because the Γ_A are real, complex conjugation of the Dirac equation gives $(-i\Gamma^A \partial_A - m)\psi^* = 0$, and because ω is real and anticommutes with every Γ_A,

$$\psi^c = \omega\, \psi^* : \qquad \big(i\Gamma^A \partial_A - m \big)\, \omega \psi^* = \omega\, \big(-i\Gamma^A \partial_A - m \big) \psi^* = 0.$$

For the quantized field $\psi^c = \omega\, (\psi^\dagger)^{\mathsf{T}}$ exchanges particles and antiparticles; in momentum space

$$(\not{p} + m)\, \omega\, u_s(\vec{p}) = 0, \qquad \omega\, u_s(\vec{p}) = \sum_{r=1}^{8} v_r(\vec{p}) \Big(-\frac{M(\vec{p})^{\mathsf{T}}}{E} \Big)_{rs}.$$

Applied twice, $(\psi^c)^c = \omega\omega\psi = -\psi$: the Majorana condition $\psi^c = \psi$ has no non-zero solution with 16 components. (A real MASSIVE spinor field in 7+1 dimensions needs 32 real components, the quaternionic module $\mathbb{H}^8 = \mathbb{R}^{32}$ of $Cl(7,1) \cong \mathbb{H}(8)$; real massless fields exist with 16 components, because $\Gamma^A \partial_A \psi = 0$ is a real equation.)

6.4 Lorentz covariance

The generators $\Sigma_{AB} = \frac{1}{4}(\Gamma_A\Gamma_B - \Gamma_B\Gamma_A)$ of Spin(1,7) are real. A boost along x_1 with rapidity $\beta_{\rm b}$ is

$$\mathcal{B}(\beta_{\rm b}) = e^{\beta_{\rm b}\Sigma_{01}} = \cosh\tfrac{\beta_{\rm b}}{2}\,1_{16} + \sinh\tfrac{\beta_{\rm b}}{2}\,\Gamma_0\Gamma_1,$$
$$\mathcal{B}\,\Gamma_K\,\mathcal{B}^{-1} = \sum_B L_{BK}\,\Gamma_B, \qquad L_{BK} = \tfrac{1}{16}\mathrm{tr}\big(\eta_{BB}\Gamma_B\,\mathcal{B}\,\Gamma_K\,\mathcal{B}^{-1}\big),$$

$$L = \begin{pmatrix} \cosh\beta_{\rm b} & -\sinh\beta_{\rm b} & 0 \\ -\sinh\beta_{\rm b} & \cosh\beta_{\rm b} & 0 \\ 0 & 0 & 1_6 \end{pmatrix}, \qquad L^{\mathsf{T}}\eta L = \eta.$$

If $\psi(x)$ solves the Dirac equation, so does $\psi'(x) = \mathcal{B}\,\psi(L^{-1}x)$, and $\mathcal{B}\,u_s(\vec{p})$ is a positive-energy spinor of the boosted momentum $p' = Lp$. Example checked in 7+1.nb: $\beta_{\rm b} = 2\log 2$ ($\cosh\frac{\beta_{\rm b}}{2} = \frac{5}{4}$, $\sinh\frac{\beta_{\rm b}}{2} = \frac{3}{4}$), $p = (5,2,2,4,0,0,0,0)$, $m = 1$:

$$p' = Lp = \Big(\frac{55}{8}, -\frac{41}{8}, 2, 4, 0, 0, 0, 0\Big), \qquad p'\cdot p' = 1, \qquad (\not{p}' - m)\,\mathcal{B}\,u_s(\vec{p}) = 0 \quad (s = 1,\ldots,8).$$

7. Chirality and the two coupled 8-component equations

7.1 No real chirality matrix in 7+1 dimensions

A chirality matrix must anticommute with every Dirac matrix and square to 1_{16}. The REAL matrices Z with $Z\Gamma_A = -\Gamma_A Z$ for $A = 0,\ldots,7$ are exactly the multiples of ω (a 255-rank linear system for the 256 entries of Z, solved in 7+1.nb), and $(c\,\omega)^2 = -c^2\,1_{16}$. Hence there is no real chirality matrix; chirality needs

$$\chi = i\,\omega = \begin{pmatrix} 0 & -i\,1_8 \\ i\,1_8 & 0 \end{pmatrix}, \qquad \chi^\dagger = \chi, \qquad \chi^2 = 1_{16}, \qquad \chi\,\Gamma_A = -\Gamma_A\,\chi, \qquad \chi\,\Sigma_{AB} = \Sigma_{AB}\,\chi,$$

so chirality is Lorentz invariant. The chiral projections are complex (the Dirac matrices stay real):

$$P_L = \tfrac{1}{2}\big(1_{16} - \chi\big), \qquad P_R = \tfrac{1}{2}\big(1_{16} + \chi\big), \qquad P_L + P_R = 1_{16}, \quad P_L^2 = P_L, \quad P_R^2 = P_R, \quad P_L P_R = 0,$$

$$\Gamma_A\,P_L = P_R\,\Gamma_A, \qquad P_L^\dagger\,\Gamma^0 = \Gamma^0\,P_R, \qquad \mathrm{rank}\,P_L = \mathrm{rank}\,P_R = 8.$$

(Mathematica: $\chi =$ `χ17`, $P_L =$ `PL17`, $P_R =$ `PR17`.)

7.2 Chiral components

The eigenvectors of χ give two 16×8 isometries (`EL`, `ER`) and the unitary matrix W (`Wch`):

$$E_L = \frac{1}{\sqrt{2}}\begin{pmatrix} 1_8 \\ -i\,1_8 \end{pmatrix}, \quad E_R = \frac{1}{\sqrt{2}}\begin{pmatrix} -i\,1_8 \\ 1_8 \end{pmatrix}, \qquad \chi E_L = -E_L, \quad \chi E_R = E_R,$$

$$W = \big(E_L, E_R\big), \qquad W^\dagger W = 1_{16},$$
$$E_L^\dagger E_L = E_R^\dagger E_R = 1_8, \qquad E_L^\dagger E_R = 0, \qquad E_L E_L^\dagger = P_L, \qquad E_R E_R^\dagger = P_R.$$

The chiral (Weyl) components of the field are the 8-component fields

$$\phi_L = E_L^\dagger\psi = \frac{\psi_\uparrow + i\,\psi_\downarrow}{\sqrt{2}}, \qquad \phi_R = E_R^\dagger\psi = \frac{i\,\psi_\uparrow + \psi_\downarrow}{\sqrt{2}},$$
$$P_L\psi = E_L\phi_L, \qquad P_R\psi = E_R\phi_R, \qquad \psi = E_L\phi_L + E_R\phi_R.$$

7.3 The chiral form of the Dirac equation

The blocks of the Dirac matrices between the chiral subspaces are

$$E_L^\dagger \beta E_L = E_R^\dagger \beta E_R = 0, \quad E_L^\dagger \beta E_R = E_R^\dagger \beta E_L = 1_8, \qquad E_L^\dagger \alpha_i E_L = -\sigma_i, \quad E_R^\dagger \alpha_i E_R = \sigma_i, \quad E_L^\dagger \alpha_i E_R = 0,$$

with $\sigma_i = i\tau_i^O$, so W brings $h(\vec{p})$ to Weyl form:

$$W^\dagger h(\vec{p})\, W = \begin{pmatrix} -\sigma\cdot\vec{p} & m\,1_8 \\ m\,1_8 & \sigma\cdot\vec{p} \end{pmatrix}.$$

Projecting the Dirac equation with $E_L^\dagger\Gamma^0$ and $E_R^\dagger\Gamma^0$ gives, for arbitrary ψ,

$$E_L^\dagger\,\Gamma^0\big(i\Gamma^A\partial_A - m\big)\psi = \big(i\partial_0 + \tau_i^O\partial_i\big)\phi_L - m\,\phi_R,$$
$$E_R^\dagger\,\Gamma^0\big(i\Gamma^A\partial_A - m\big)\psi = \big(i\partial_0 - \tau_i^O\partial_i\big)\phi_R - m\,\phi_L,$$

so the 16-component Dirac equation is equivalent to the two coupled 8-component equations

$$\big(i\,\partial_0 + \tau_i^O\,\partial_i\big)\,\phi_L = m\,\phi_R,$$
$$\big(i\,\partial_0 - \tau_i^O\,\partial_i\big)\,\phi_R = m\,\phi_L.$$

Only the REAL 8×8 matrices τ_i^O occur (because $\sigma_i(-i\partial_i) = \tau_i^O\partial_i$); the only i is the one in $i\partial_0$. With $\sigma^A = (1_8, \sigma_i)$ and $\bar\sigma^A = (1_8, -\sigma_i)$ the covariant form is

$$i\,\bar\sigma^A\partial_A\,\phi_L = m\,\phi_R, \qquad i\,\sigma^A\partial_A\,\phi_R = m\,\phi_L.$$

The mass couples the two chiralities; for $m = 0$ the equations decouple into two independent 8-component Weyl equations. Each 8-component field obeys the Klein–Gordon equation, because

$$\big(i\partial_0 - \tau_i^O\partial_i\big)\big(i\partial_0 + \tau_j^O\partial_j\big) = -\partial_0^2 - \tau_i^O\tau_j^O\partial_i\partial_j = -\Box\,1_8.$$

For a plane wave $e^{-i(Ex_0 - \vec{p}\cdot\vec{x})}$ ($i\partial_0 \to E$, $\tau_i^O\partial_i \to i\,\tau^O\cdot\vec{p} = \sigma\cdot\vec{p}$) the chiral equations read $(E+\sigma\cdot\vec{p})\phi_L = m\phi_R$ and $(E - \sigma\cdot\vec{p})\phi_R = m\phi_L$; 7+1.nb checks them for the eight spinors u_s of the example of section 5.2.

7.4 Bilinears in chiral form

With $\kappa = \phi_L^\dagger\phi_R$ (a complex scalar):

$$\bar\psi\psi = \phi_L^\dagger\phi_R + \phi_R^\dagger\phi_L = \kappa + \kappa^*, \qquad \bar\psi\, i\chi\,\psi = i\big(\kappa - \kappa^*\big),$$
$$\bar\psi\Gamma^0\psi = \phi_L^\dagger\phi_L + \phi_R^\dagger\phi_R, \qquad \bar\psi\Gamma^i\psi = -\phi_L^\dagger\sigma_i\phi_L + \phi_R^\dagger\sigma_i\phi_R,$$
$$\bar\psi\Gamma^0\chi\psi = -\phi_L^\dagger\phi_L + \phi_R^\dagger\phi_R, \qquad \bar\psi\Gamma^i\chi\psi = \phi_L^\dagger\sigma_i\phi_L + \phi_R^\dagger\sigma_i\phi_R,$$
$$E_L^\dagger\beta\, i\chi\, E_R = i\,1_8, \quad E_R^\dagger\beta\, i\chi\, E_L = -i\,1_8, \quad E_L^\dagger\chi E_L = -1_8, \quad E_R^\dagger\chi E_R = 1_8, \quad E_{L,R}^\dagger\alpha_i\chi E_{L,R} = \sigma_i.$$

8. The self-coupled field and its 16-component Kohn–Sham theory

8.1 Lagrangian, densities and Hamiltonian

The field couples to itself through the four Lorentz-invariant contact (point) interactions built from bilinears without derivatives, and to four real external c-number sources $\mathsf{s}(x)$, $\mathsf{p}(x)$, $\mathsf{v}_A(x)$, $\mathsf{a}_A(x)$:

$$\mathcal{L} = \bar\psi\big(i\Gamma^A\partial_A - m\big)\psi - \bar\psi\big(\mathsf{s} + i\chi\,\mathsf{p} + \Gamma^A(\mathsf{v}_A + \chi\,\mathsf{a}_A)\big)\psi$$
$$- \frac{g_S}{2}\big(\bar\psi\psi\big)^2 - \frac{g_P}{2}\big(\bar\psi\, i\chi\,\psi\big)^2 - \frac{g_V}{2}\big(\bar\psi\Gamma^A\psi\big)\big(\bar\psi\Gamma_A\psi\big) - \frac{g_A}{2}\big(\bar\psi\Gamma^A\chi\psi\big)\big(\bar\psi\Gamma_A\chi\psi\big).$$

The four densities are

$$\rho_S = \bar\psi\psi \ \text{ (scalar)}, \qquad \rho_P = \bar\psi\, i\chi\,\psi \ \text{ (pseudoscalar)},$$

$$j^A = \bar\psi\,\Gamma^A\psi \;\;\text{(vector)}, \qquad j_5^A = \bar\psi\,\Gamma^A\chi\,\psi \;\;\text{(axial vector)},$$

with the charge density $j^0 = \psi^\dagger\psi$; a lower index is $j_A = \eta_{AA}j^A$, so $j^A j_A = (j^0)^2 - \sum_i (j^i)^2$. All four are real (Hermitian operators), because $\Gamma^0 O$ is a Hermitian matrix for $O = 1_{16},\ i\chi,\ \Gamma^A,\ \Gamma^A\chi$. The couplings g_S, g_P, g_V, g_A are real constants. The interaction has no time derivatives, so it enters the Hamiltonian density with the opposite sign (normal ordered):

$$\begin{aligned}\mathcal{H} =& :\psi^\dagger\big(-i\alpha_i\partial_i + \beta m\big)\psi: + :\bar\psi\big(\mathsf{s} + i\chi\,\mathsf{p} + \Gamma^A(\mathsf{v}_A + \chi\,\mathsf{a}_A)\big)\psi: \\ &+ \frac{g_S}{2} :\rho_S^2: + \frac{g_P}{2} :\rho_P^2: + \frac{g_V}{2} : j^A j_A : + \frac{g_A}{2} : j_5^A j_{5A} : .\end{aligned}$$

Hence $g_S < 0$ is attractive (it lowers the effective mass) and $g_V > 0$ is repulsive for the charge density. **Dimensions.** With $\hbar = c = 1$ the action is dimensionless; in 7+1 dimensions ψ has mass dimension $7/2$ and every coupling g has mass dimension -6: the contact couplings are not renormalizable. The model is an effective theory. It needs a momentum cutoff Λ when the Dirac sea is included (section 10.4), or the no-sea approximation of relativistic mean-field and point-coupling models (section 8.4).

8.2 The nonlinear field equation

The Euler–Lagrange equation $\partial^L\mathcal{L}/\partial\bar\psi_a = 0$ with the Grassmann LEFT derivative (the bilinears are Grassmann-even, so moving $\bar\psi_a$ to the front produces no other signs) is the nonlinear Dirac equation

$$\begin{aligned}&\big(i\Gamma^A\partial_A - m - \mathsf{s} - i\chi\,\mathsf{p} - \Gamma^A(\mathsf{v}_A + \chi\,\mathsf{a}_A)\big)\psi \\ &\quad = g_S\big(\bar\psi\psi\big)\psi + g_P\big(\bar\psi\, i\chi\psi\big)\, i\chi\psi \\ &\qquad + g_V\big(\bar\psi\Gamma_A\psi\big)\Gamma^A\psi + g_A\big(\bar\psi\Gamma_A\chi\psi\big)\Gamma^A\chi\psi.\end{aligned}$$

7+1.nb verifies it by differentiating the interaction Lagrangian in a Grassmann algebra with 32 generators ($\psi_a = \theta_a$, $\bar\psi_a = \theta_{16+a}$, $a = 1,\dots,16$): for every a

$$\frac{\partial^L\mathcal{L}_{\rm int}}{\partial\bar\psi_a} = -\Big[g_S(\bar\psi\psi)\psi + g_P(\bar\psi\, i\chi\psi)\, i\chi\psi + g_V(\bar\psi\Gamma_A\psi)\Gamma^A\psi + g_A(\bar\psi\Gamma_A\chi\psi)\Gamma^A\chi\psi\Big]_a .$$

8.3 Densities, sources and the Hohenberg–Kohn theorem

Collect the densities into $n = (\rho_S, \rho_P, j^A, j_5^A)$ — $1+1+8+8 = 18$ real fields on space — and the static sources into $w = (\mathsf{s}, \mathsf{p}, \mathsf{v}_A, \mathsf{a}_A)$:

$$H[w] = H_0 + H_{\rm int} + \int w{\cdot}n\, d^7x, \qquad w{\cdot}n = \mathsf{s}\,\rho_S + \mathsf{p}\,\rho_P + \mathsf{v}_A\, j^A + \mathsf{a}_A\, j_5^A, \qquad H_0 = \int :\psi^\dagger\big(-i\alpha_i\partial_i + \beta m\big)\psi:\, d^7x.$$

The charge $Q = \int j^0 d^7x$ commutes with $H[w]$; the theory is formulated in the sector of fixed charge $Q = N$.

Theorem (Hohenberg–Kohn, relativistic form). Let $|\Psi\rangle$ be the non-degenerate ground state of $H[w]$ with charge N and densities n. If a second source w' with a different ground state $|\Psi'\rangle$ had the same densities n, the variational principle would give

$$E = \langle\Psi|H[w]|\Psi\rangle < \langle\Psi'|H[w]|\Psi'\rangle = E' + \int (w - w')\cdot n\, d^7x, \qquad E' < E + \int (w' - w)\cdot n\, d^7x,$$

and adding the two inequalities gives $E + E' < E + E'$, a contradiction. Hence the ground-state densities determine the ground state and every ground-state property; they determine w up to changes that leave the ground state unchanged (for example $\mathsf{v}_0 \to \mathsf{v}_0 + \text{const}$, which only adds $\text{const}\times N$ to the energy). The energy is

$$E[w] = \min_n \Big(F[n] + \int w\cdot n\, d^7x\Big), \qquad F[n] = \min_{|\Psi\rangle\to n} \langle\Psi|H_0 + H_{\rm int}|\Psi\rangle,$$

with the universal (source-independent) functional F defined by the Levy–Lieb constrained search over states with densities n. Standing assumptions, as usual in relativistic density-functional theory: the theory is regularized (cutoff or no-sea), $H[w]$ is bounded below in the sector $Q = N$, and the densities considered are ground-state densities of some w (v-representability).

8.4 The Kohn–Sham equations

The Kohn–Sham reference system consists of N non-interacting 16-component fermions φ_k ($k=1,\dots,N$; $\int \varphi_k^\dagger \varphi_l \, d^7x = \delta_{kl}$) in effective sources $w_{\mathrm{KS}} = (\mathcal{S}, \mathcal{P}, \mathcal{V}_A, \mathcal{A}_A)$, chosen such that its ground state has the same densities n as the interacting system. Split

$$F[n] = T_s[n] + E_H[n] + E_{xc}[n], \qquad T_s = \sum_{k=1}^{N} \int \varphi_k^\dagger \big(-i\alpha_i\partial_i + \beta m \big)\varphi_k \, d^7x,$$

$$E_H = \int \Big[\frac{g_S}{2}\rho_S^2 + \frac{g_P}{2}\rho_P^2 + \frac{g_V}{2} j^A j_A + \frac{g_A}{2} j_5^A j_{5A} \Big] d^7x \qquad \text{(Hartree, the direct term)},$$

and $E_{xc} = F - T_s - E_H$ (exchange and correlation). Minimizing $E = T_s + \int w\cdot n + E_H + E_{xc}$ over orthonormal orbitals (Lagrange multipliers ε_k) gives the Kohn–Sham equations

$$\begin{aligned}
h_{\mathrm{KS}}\,\varphi_k &= \varepsilon_k\,\varphi_k, \\
h_{\mathrm{KS}} &= \alpha_i\big(-i\partial_i + \mathcal{V}_i\big) + \beta\big(m + \mathcal{S}\big) + \mathcal{V}_0 + \beta\, i\chi\,\mathcal{P} + \chi\,\mathcal{A}_0 + \alpha_i\chi\,\mathcal{A}_i, \\
\mathcal{S} &= \mathsf{s} + g_S\,\rho_S + \frac{\delta E_{xc}}{\delta \rho_S}, \qquad \mathcal{P} = \mathsf{p} + g_P\,\rho_P + \frac{\delta E_{xc}}{\delta \rho_P}, \\
\mathcal{V}_A &= \mathsf{v}_A + g_V\, j_A + \frac{\delta E_{xc}}{\delta j^A}, \qquad \mathcal{A}_A = \mathsf{a}_A + g_A\, j_{5A} + \frac{\delta E_{xc}}{\delta j_5^A}, \\
\rho_S &= \sum_k \bar\varphi_k\varphi_k, \quad \rho_P = \sum_k \bar\varphi_k\, i\chi\,\varphi_k, \quad j^A = \sum_k \bar\varphi_k\Gamma^A\varphi_k, \quad j_5^A = \sum_k \bar\varphi_k\Gamma^A\chi\,\varphi_k.
\end{aligned}$$

$\mathcal{V}_A$ and $\mathcal{A}_A$ carry a LOWER index ($\mathcal{V}_i = -\mathcal{V}^i$). Equivalently every orbital $\varphi_k\, e^{-i\varepsilon_k x_0}$ solves the Dirac equation with the self-consistent potentials,

$$\big(i\Gamma^A\partial_A - m - \mathcal{S} - i\chi\,\mathcal{P} - \Gamma^A(\mathcal{V}_A + \chi\,\mathcal{A}_A)\big)\varphi = 0, \qquad \Gamma^0\big(\not{p} - m - \mathcal{S} - i\chi\mathcal{P} - \Gamma^A(\mathcal{V}_A + \chi\mathcal{A}_A)\big) = \varepsilon\,1_{16} - h_{\mathrm{KS}}(\vec p)$$

for a plane wave with energy ε ($\not{p} = \varepsilon\Gamma_0 + p^i\Gamma_i$). h_{KS} is Hermitian for real potentials, because α_i, β, $\beta i\chi$, χ and $\alpha_i\chi$ are Hermitian.

Why these are the potentials. With φ and φ^* treated as independent variables (Wirtinger calculus), the Hartree energy density $e_H(\varphi,\varphi^*)$ of one orbital satisfies, for every component a,

$$\frac{\partial e_H}{\partial \varphi_a^*} = \big(v_H\,\varphi\big)_a, \qquad v_H = \beta\, g_S\rho_S + \beta\, i\chi\, g_P\rho_P + \sum_A \Gamma^0\Gamma^A\, g_V j_A + \sum_A \Gamma^0\Gamma^A\chi\, g_A j_{5A},$$

i.e. v_H is $h_{\mathrm{KS}} - h$ with $\mathcal{S} = g_S\rho_S$, $\mathcal{P} = g_P\rho_P$, $\mathcal{V}_A = g_V j_A$, $\mathcal{A}_A = g_A j_{5A}$ (checked symbolically in 7+1.nb with 32 independent variables).

No-sea approximation. The occupied orbitals are the N lowest POSITIVE-energy eigenvectors of h_{KS}; the Dirac sea is not included and its effects are absorbed in the couplings, as in relativistic mean-field and point-coupling models of nuclei.

Total energy.

$$E = \sum_{k=1}^{N}\varepsilon_k - \int \Big[(\mathcal{S}-\mathsf{s})\rho_S + (\mathcal{P}-\mathsf{p})\rho_P + (\mathcal{V}_A - \mathsf{v}_A)j^A + (\mathcal{A}_A - \mathsf{a}_A)j_5^A\Big] d^7x + E_H + E_{xc};$$

in the Hartree approximation ($E_{xc}=0$) simply $E = \sum_k \varepsilon_k - E_H$ (double counting). **Self-consistency loop:** guess the potentials; solve $h_{\mathrm{KS}}\varphi_k = \varepsilon_k\varphi_k$; occupy the N lowest positive levels; compute n; compute the new potentials; repeat until they no longer change.

8.5 Exchange: Wick's theorem for contact couplings

For a Slater determinant $|\Phi\rangle$ of N orthonormal orbitals (no sea) and two vertex matrices O_1, O_2, all fields at the same point (contact interaction), Wick's theorem gives

$$\langle\Phi| : \big(\bar\psi O_1\psi\big)\big(\bar\psi O_2\psi\big) : |\Phi\rangle = \mathrm{tr}(O_1R)\,\mathrm{tr}(O_2R) - \mathrm{tr}(O_1RO_2R), \qquad R = \sum_{k=1}^{N}\varphi_k\,\bar\varphi_k \quad (16\times 16).$$

Derivation: with mode operators c_a at the point, $:c_a^\dagger c_b c_c^\dagger c_d: = c_a^\dagger c_c^\dagger c_d c_b$ and $\langle c_a^\dagger c_c^\dagger c_d c_b\rangle = \varrho_{ba}\varrho_{dc} - \varrho_{da}\varrho_{bc}$ with $\varrho_{xy} = \langle c_y^\dagger c_x\rangle = \sum_k \varphi_{k,x}\varphi_{k,y}^*$; contracting with $(\Gamma^0 O_1)_{ab}(\Gamma^0 O_2)_{cd}$ and using $R = \varrho\,\Gamma^0$ gives the formula. The first term is the Hartree (direct) energy, the second the exchange (Fock) term; its minus sign comes from the anticommuting fields. 7+1.nb checks the formula for all 18 vertex pairs (1 scalar, 1 pseudoscalar, 8 vector, 8 axial) in the 3-particle Fock space of 16 modes (dimension $\binom{16}{3} = 560$), where every $c_a^\dagger c_b$ is an explicit sparse matrix and

$$:\big(\psi^\dagger M_1\psi\big)\big(\psi^\dagger M_2\psi\big): = B(M_1)\,B(M_2) - B(M_1M_2), \qquad B(M) = \sum_{a,b} M_{ab}\,c_a^\dagger c_b, \qquad M_{1,2} = \Gamma^0 O_{1,2}.$$

The exact exchange energy of the model is therefore an explicit functional of the occupied orbitals,

$$E_x = -\frac{g_S}{2}\mathrm{tr}(RR) - \frac{g_P}{2}\mathrm{tr}(i\chi R\, i\chi R) - \frac{g_V}{2}\sum_A \mathrm{tr}\big(\Gamma^A R\,\Gamma_A R\big) - \frac{g_A}{2}\sum_A \mathrm{tr}\big(\Gamma^A \chi R\,\Gamma_A \chi R\big),$$

and its variation $\delta E_x = \mathrm{tr}(\Sigma_x\,\delta R)$ defines the exchange self-energy

$$\Sigma_x = -g_S\,R - g_P\,(i\chi)R(i\chi) - g_V\sum_A \Gamma_A R\,\Gamma^A - g_A\sum_A \Gamma_A\chi R\,\Gamma^A\chi, \qquad E_x = \tfrac{1}{2}\,\mathrm{tr}\big(\Sigma_x R\big).$$

In the orbital equations it enters as $\big(i\Gamma^A\partial_A - m - \Sigma_H - \Sigma_x\big)\varphi = 0$ with the Hartree part $\Sigma_H = \mathcal{S} + i\chi\mathcal{P} + \Gamma^A(\mathcal{V}_A + \chi\mathcal{A}_A)$. The hierarchy of approximations is: $E_{xc} = 0$ (relativistic Hartree, mean field); $E_{xc} = E_x$ (relativistic Hartree–Fock, exact exchange); $E_{xc} = E_x + E_c$ with a model for the correlation energy E_c.

8.6 Fierz identities

The 256 products Γ_K (section 3.1) are orthogonal matrices, $\Gamma_K^{-1} = \Gamma_K^{\mathsf{T}}$, with

$$\mathrm{tr}\big(\Gamma_K^{-1}\Gamma_L\big) = 16\,\delta_{KL}, \qquad \sum_K \big(\Gamma_K\big)_{ab}\big(\Gamma_K^{-1}\big)_{cd} = 16\,\delta_{ad}\,\delta_{cb} \qquad \text{(Fierz completeness)}.$$

Consequently every exchange term is a sum of direct terms (Fierz rearrangement),

$$\mathrm{tr}\big(O_1 R O_2 R\big) = \frac{1}{16}\sum_K \mathrm{tr}\big(\Gamma_K^{-1}R\big)\,\mathrm{tr}\big(O_1\Gamma_K O_2 R\big),$$

so for contact couplings Hartree–Fock is a Hartree theory with Fierz-transformed couplings (section 10.2 shows it for uniform matter). 7+1.nb checks these identities with all 256 products and a random complex matrix R.

9. The chiral Kohn–Sham theory (8 + 8 components)

9.1 Chiral densities and sources

With the chiral components $\varphi_{kL} = E_L^\dagger\varphi_k$ and $\varphi_{kR} = E_R^\dagger\varphi_k$ of the Kohn–Sham orbitals (section 7.2) the 18 densities become one complex scalar density and two 8-currents:

$$\kappa = \sum_k \varphi_{kL}^\dagger\varphi_{kR}, \qquad j_L^A = \sum_k \varphi_{kL}^\dagger\bar\sigma^A\varphi_{kL}, \qquad j_R^A = \sum_k \varphi_{kR}^\dagger\sigma^A\varphi_{kR},$$
$$\rho_S = \kappa + \kappa^* = 2\,\mathrm{Re}\,\kappa, \qquad \rho_P = i\big(\kappa - \kappa^*\big) = -2\,\mathrm{Im}\,\kappa, \qquad j^A = j_L^A + j_R^A, \qquad j_5^A = j_R^A - j_L^A,$$

with $\sigma^A = (1_8, \sigma_i)$ and $\bar\sigma^A = (1_8, -\sigma_i)$. The scalar and pseudoscalar densities pair the two chiralities; the currents do not. The sources become one complex scalar source $\varpi = \mathsf{s} + i\mathsf{p}$ and two chiral vector sources:

$$\mathsf{s}\,\rho_S + \mathsf{p}\,\rho_P = \varpi\,\kappa + \varpi^*\kappa^*, \qquad \mathsf{v}_A j^A + \mathsf{a}_A j_5^A = (\mathsf{v}-\mathsf{a})_A\,j_L^A + (\mathsf{v}+\mathsf{a})_A\,j_R^A.$$

9.2 The Hartree energy in chiral variables

$$\begin{aligned} e_H &= \frac{g_S}{2}\big(\kappa+\kappa^*\big)^2 - \frac{g_P}{2}\big(\kappa-\kappa^*\big)^2 + \frac{g_V+g_A}{2}\big(j_L^A j_{LA} + j_R^A j_{RA}\big) + (g_V-g_A)\, j_L^A j_{RA} \\ &= \frac{g_S-g_P}{2}\big(\kappa^2+\kappa^{*2}\big) + (g_S+g_P)\,\kappa\kappa^* + \frac{g_V+g_A}{2}\big(j_L^A j_{LA} + j_R^A j_{RA}\big) + (g_V-g_A)\, j_L^A j_{RA}. \end{aligned}$$

The scalar–pseudoscalar part depends on κ only through $|\kappa|^2$ exactly when $g_P = g_S$; then it is $2g_S|\kappa|^2$, the Nambu–Jona-Lasinio form. The vector–axial part couples the two currents only through $g_V - g_A$.

9.3 The chiral Kohn–Sham equations

In chiral components, $W^\dagger h_{\rm KS} W$, the Kohn–Sham equation becomes the 8+8 block system

$$\begin{aligned} \big[\mathcal{V}_0^L - \sigma_i\big(-i\partial_i + \mathcal{V}_i^L\big)\big]\varphi_L + \mathcal{M}\,\varphi_R &= \varepsilon\,\varphi_L, \\ \mathcal{M}^*\,\varphi_L + \big[\mathcal{V}_0^R + \sigma_i\big(-i\partial_i + \mathcal{V}_i^R\big)\big]\varphi_R &= \varepsilon\,\varphi_R, \end{aligned}$$

$$\mathcal{M} = m + \mathcal{S} + i\,\mathcal{P} \quad \text{(complex Kohn–Sham mass)}, \qquad \mathcal{V}_A^L = \mathcal{V}_A - \mathcal{A}_A, \qquad \mathcal{V}_A^R = \mathcal{V}_A + \mathcal{A}_A,$$

$$W^\dagger h_{\rm KS}(\vec{p})\, W = \begin{pmatrix} \mathcal{V}_0^L - \sigma_i(p^i + \mathcal{V}_i^L) & \mathcal{M}\,1_8 \\ \mathcal{M}^*\,1_8 & \mathcal{V}_0^R + \sigma_i(p^i + \mathcal{V}_i^R) \end{pmatrix}.$$

Since $\sigma_i(-i\partial_i) = \tau_i^O \partial_i$, the derivative terms contain only the real τ_i^O, exactly as in section 7.3; the complex mass $\mathcal{M}$ couples the chiralities. The potentials are Wirtinger derivatives of the energy (κ and κ^* independent):

$$\mathcal{M} - m = \varpi + \frac{\partial(e_H + e_{xc})}{\partial\kappa}, \qquad \mathcal{V}_A^L = (\mathsf{v}-\mathsf{a})_A + \frac{\partial(e_H+e_{xc})}{\partial j_L^A}, \qquad \mathcal{V}_A^R = (\mathsf{v}+\mathsf{a})_A + \frac{\partial(e_H+e_{xc})}{\partial j_R^A};$$

in the Hartree approximation

$$\begin{aligned} \mathcal{M} - m - \varpi &= g_S\big(\kappa+\kappa^*\big) - g_P\big(\kappa-\kappa^*\big) = g_S\,\rho_S + i\,g_P\,\rho_P, \\ \mathcal{V}_A^L - (\mathsf{v}-\mathsf{a})_A &= (g_V+g_A)\, j_{LA} + (g_V-g_A)\, j_{RA}, \qquad \mathcal{V}_A^R - (\mathsf{v}+\mathsf{a})_A = (g_V+g_A)\, j_{RA} + (g_V-g_A)\, j_{LA}, \end{aligned}$$

and, per orbital, $\partial e_H/\partial\varphi_L^* = (\mathcal{V}_0^L - \sigma_i\mathcal{V}_i^L)\varphi_L + (\mathcal{M}-m)\varphi_R$, $\partial e_H/\partial\varphi_R^* = (\mathcal{V}_0^R + \sigma_i\mathcal{V}_i^R)\varphi_R + (\mathcal{M}-m)^*\varphi_L$ (all verified symbolically in 7+1.nb). The densities κ, j_L, j_R are computed from the occupied chiral orbitals (no sea, as in section 8.4), and

$$E = \sum_k \varepsilon_k - \int \Big[2\,\mathrm{Re}\big((\mathcal{M}-m-\varpi)\kappa\big) + \big(\mathcal{V}^L - \mathsf{v} + \mathsf{a}\big)_A j_L^A + \big(\mathcal{V}^R - \mathsf{v} - \mathsf{a}\big)_A j_R^A\Big] d^7x + E_H + E_{xc}.$$

9.4 Chiral symmetry

Because $\chi = i\omega$, the chiral transformation is a REAL rotation of the 16 components:

$$\psi \to e^{i\theta\chi}\psi = \big(\cos\theta\, 1_{16} - \sin\theta\,\omega\big)\psi, \qquad \varphi_L \to e^{-i\theta}\varphi_L, \quad \varphi_R \to e^{i\theta}\varphi_R, \quad \kappa \to e^{2i\theta}\kappa.$$

The currents j_L, j_R (hence j and j_5) are invariant, $\big(e^{i\theta\chi}\big)^\dagger \beta\, e^{i\theta\chi} = \beta\,(\cos 2\theta + i\sin 2\theta\,\chi)$, so (ρ_S, ρ_P) rotate by the angle 2θ, and

$$E_H\big(e^{i\theta\chi}\psi\big) - E_H(\psi) = \frac{g_S-g_P}{2}\Big[\big(e^{4i\theta}-1\big)\kappa^2 + \big(e^{-4i\theta}-1\big)\kappa^{*2}\Big]:$$

the interaction is chirally $(U(1)_A)$ symmetric if and only if $g_P = g_S$ (checked in 7+1.nb with $\cos\theta = \frac{3}{5}$, $\sin\theta = \frac{4}{5}$). The mass term $m(\kappa+\kappa^*)$ of T_s breaks the symmetry explicitly. For $m = 0$, $g_P = g_S$ and $\varpi = 0$ the whole chiral Kohn–Sham theory is invariant: with $\kappa = 0$ the chiral equations decouple into separate 8-component equations for φ_L and φ_R, while a self-consistent solution with $\kappa \neq 0$ breaks the symmetry spontaneously (section 10.4).

10. Worked examples

10.1 Uniform matter in the Hartree approximation (no sea)

Static, uniform, spin-saturated matter at rest, without external sources. By symmetry the Kohn–Sham potentials are constants, $\mathcal{S} = m^* - m$ and $\mathcal{V}_0$, with $\mathcal{P} = \mathcal{A}_A = \mathcal{V}_i = 0$. The orbitals are plane waves $u_s(\vec{p})\, e^{i\vec{p}\cdot\vec{x}}/\sqrt{2E^*V}$ with the effective mass m^*, $E^* = \sqrt{\vec{p}^2 + m^{*2}}$, eigenvalues $\varepsilon = \mathcal{V}_0 + E^*$, and all 8 spin states filled up to the Fermi momentum k_F. The unit sphere $S^6 \subset \mathbb{R}^7$ has area $|S^6| = 2\pi^{7/2}/\Gamma_{\rm E}(\frac{7}{2}) = 16\pi^3/15$ ($\Gamma_{\rm E}$ is Euler's Gamma function, $\Gamma_{\rm E}(\frac{7}{2}) = \frac{15}{8}\sqrt{\pi}$), so with 8 spin states

$$8\int \frac{d^7k}{(2\pi)^7}\, f(|\vec{k}|) = c_7 \int_0^\infty k^6 f(k)\, dk, \qquad c_7 = \frac{8\,|S^6|}{(2\pi)^7} = \frac{1}{15\pi^4},$$

$$\rho = j^0 = c_7 \frac{k_F^7}{7} = \frac{k_F^7}{105\pi^4}, \qquad \rho_S = c_7 \int_0^{k_F} \frac{k^6\, m^*}{\sqrt{k^2 + m^{*2}}}\, dk, \qquad \varepsilon_{\rm kin} = c_7 \int_0^{k_F} k^6 \sqrt{k^2 + m^{*2}}\, dk,$$

($\rho_P = 0$, $j_5 = 0$, $j^i = 0$). The occupied plane waves give the density matrix $R = \big(\rho_S\, 1_{16} + \rho\, \Gamma_0\big)/16$, because $\frac{1}{2}\sum_s \big[u_s(\vec{p})\bar{u}_s(\vec{p}) + u_s(-\vec{p})\bar{u}_s(-\vec{p})\big]/(2E) = (E\Gamma_0 + m)/(2E)$. The integrals have the closed forms (printed by 7+1.nb and checked by differentiation)

$$\int_0^{k_F} \frac{k^6\, m^*}{\sqrt{k^2 + m^{*2}}}\, dk = \frac{m^*}{48}\Big[k_F\sqrt{k_F^2 + m^{*2}}\,\big(8k_F^4 - 10k_F^2 m^{*2} + 15m^{*4}\big) - 15\, m^{*6}\,\mathrm{artanh}\frac{k_F}{\sqrt{k_F^2 + m^{*2}}}\Big],$$

$$\int_0^{k_F} k^6\sqrt{k^2 + m^{*2}}\, dk = \frac{1}{384}\Big[k_F\sqrt{k_F^2 + m^{*2}}\,\big(48k_F^6 + 8k_F^4 m^{*2} - 10k_F^2 m^{*4} + 15m^{*6}\big) - 15\, m^{*8}\,\mathrm{artanh}\frac{k_F}{\sqrt{k_F^2 + m^{*2}}}\Big].$$

The Hartree self-consistency conditions, the energy density and the Fermi energy are

$$m^* = m + g_S\, \rho_S(m^*), \qquad \mathcal{V}_0 = g_V\, \rho,$$
$$\varepsilon = \varepsilon_{\rm kin}(m^*) - (m^* - m)\,\rho_S + \frac{g_S}{2}\rho_S^2 + \frac{g_V}{2}\rho^2 = \varepsilon_{\rm kin}(m^*) - \frac{(m^* - m)^2}{2g_S} + \frac{g_V}{2}\rho^2,$$
$$\varepsilon_F = \mathcal{V}_0 + \sqrt{k_F^2 + m^{*2}}.$$

The energy density equals the sum of the occupied eigenvalues minus the double counting, $\varepsilon = \mathcal{V}_0\rho + \varepsilon_{\rm kin} - \frac{g_S}{2}\rho_S^2 - \frac{g_V}{2}\rho^2$. It is stationary in m^*, $\partial\varepsilon/\partial m^* = \rho_S - (m^* - m)/g_S = 0$ being the gap equation (using $d\varepsilon_{\rm kin}/dm^* = \rho_S$), so $d\varepsilon/dk_F = \partial\varepsilon/\partial k_F = c_7 k_F^6\,(\sqrt{k_F^2 + m^{*2}} + g_V\rho)$ and, with $d\rho/dk_F = c_7 k_F^6$, the chemical potential μ is

$$\mu = \frac{d\varepsilon}{d\rho} = \varepsilon_F \qquad \text{at every density,} \qquad \frac{d}{dk_F}\Big(\frac{\varepsilon}{\rho}\Big) = \frac{c_7 k_F^6}{\rho}\Big(\mu - \frac{\varepsilon}{\rho}\Big):$$

at the saturation point (the minimum of the energy per particle $E/A = \varepsilon/\rho$) one has $E/A = \varepsilon_F$ — the Hugenholtz–Van Hove theorem. **Example** (units $m = 1$): $g_S = -10^5$, $g_V = 10^4$, $g_P = g_A = 0$. Saturation is found by solving $\varepsilon/\rho = \varepsilon_F$ for k_F (40-digit arithmetic in 7+1.nb, 30 digits in Python):

quantity	**saturation point**	$k_F = 1$
k_F	0.756636150701493	1
ρ	$1.38810021323169 \times 10^{-5}$	$9.77712595684222 \times 10^{-5}$
m^*	0.348097340146243	0.0809644302776061
ρ_S	$6.51902659853757 \times 10^{-6}$	—
$\mathcal{V}_0 = g_V\rho$	0.138810021323169	—
$E/A - m$	−0.0283214279647386	0.410864848634332
ε_F	0.971678572035261	1.98098486130815

At the saturation point $E/A = 1 + (E/A - m) = \varepsilon_F$ (to 25 digits) and the numerical derivative $d\varepsilon/d\rho$ equals ε_F; at $k_F = 1$ the matter is not saturated ($E/A - m > 0$) and still $d\varepsilon/d\rho = \varepsilon_F$. The energy per particle is lower than at $0.95\,k_F$ and $1.05\,k_F$ of the saturation point: the matter is bound and saturates.

10.2 Uniform matter in the Hartree–Fock approximation

With the uniform density matrix $R = (\rho_S 1_{16} + \rho\Gamma_0)/16$ the exchange energy and self-energy of section 8.5 are

$$E_x = -\frac{g_S}{32}\big(\rho_S^2 + \rho^2\big) + \frac{g_P}{32}\big(\rho_S^2 - \rho^2\big) - \frac{g_V}{32}\big(8\rho_S^2 - 6\rho^2\big) + \frac{g_A}{32}\big(8\rho_S^2 + 6\rho^2\big),$$
$$\Sigma_x = S_x\, 1_{16} + V_x\, \Gamma_0, \qquad S_x = \frac{\partial E_x}{\partial \rho_S} = \frac{\rho_S}{16}\big(-g_S + g_P - 8g_V + 8g_A\big),$$
$$V_x = \frac{\partial E_x}{\partial \rho} = \frac{\rho}{16}\big(-g_S - g_P + 6g_V + 6g_A\big),$$

(using $\Gamma^A\Gamma_A = 8$ and $\Gamma^A\Gamma_0\Gamma_A = -6\Gamma_0$), so uniform Hartree–Fock is Hartree with the Fierz-transformed couplings

$$G_S = g_S + \frac{-g_S + g_P - 8g_V + 8g_A}{16}, \qquad G_V = g_V + \frac{-g_S - g_P + 6g_V + 6g_A}{16},$$
$$\frac{g_S}{2}\rho_S^2 + \frac{g_V}{2}\rho^2 + E_x = \frac{G_S}{2}\rho_S^2 + \frac{G_V}{2}\rho^2.$$

For $g_S = -10^5$, $g_V = 10^4$, $g_P = g_A = 0$: $G_S = -98750$, $G_V = 20000$. At $k_F = \frac{3}{4}$:

quantity	**Hartree–Fock**	**Hartree**
m^*	0.366436301981544	0.362973947648030
$\mathcal{V}_0$	0.261017754731616	0.130508877365808
$E/A - m$	0.0391463380161199	−0.0280653758468730

The Hartree–Fock self-consistency $m^* - m = g_S\rho_S + S_x$, $\mathcal{V}_0 = g_V\rho + V_x$ and $\mu = \varepsilon_F$ hold to 30 digits. The exchange term weakens the attraction and strengthens the repulsion: with these couplings Hartree–Fock matter at $k_F = \frac{3}{4}$ is unbound.

10.3 Uniform matter in chiral form

For the matter of section 10.1: $\kappa = \rho_S/2$ (real), $j_L^0 = j_R^0 = \rho/2$, $j_L^i = j_R^i = 0$, $\mathcal{M} = m^*$ (real), $\mathcal{V}_0^L = \mathcal{V}_0^R = \mathcal{V}_0$. The chiral Kohn–Sham Hamiltonian $\left(\begin{smallmatrix} \mathcal{V}_0 - \sigma\cdot\vec{p} & m^* \\ m^* & \mathcal{V}_0 + \sigma\cdot\vec{p}\end{smallmatrix}\right)$ satisfies $(h_\chi - \mathcal{V}_0)^2 = (\vec{p}^{\,2} + m^{*2})1_{16}$: eigenvalues $\mathcal{V}_0 \pm \sqrt{\vec{p}^{\,2} + m^{*2}}$, eight times each, as in the 16-component form. At saturation: $\kappa = 3.25951329927 \times 10^{-6}$, $j_L^0 = j_R^0 = 6.94050106616 \times 10^{-6}$, $\mathcal{M} = 0.348097340146$, $\mathcal{V}_0^L = \mathcal{V}_0^R = 0.138810021323$.

10.4 The chiral vacuum with the Dirac sea: spontaneous chiral symmetry breaking

Keep now the Dirac sea — all negative-energy Kohn–Sham orbitals occupied, with a momentum cutoff $|\vec{p}| < \Lambda$ — and take $g_P = g_S = g$, $\varpi = 0$, $\mathcal{V} = \mathcal{A} = 0$, no matter. With a uniform complex mass $\mathcal{M}$ the chiral Hamiltonian

$$h_\chi(\vec{p}) = \begin{pmatrix} -\sigma\cdot\vec{p} & \mathcal{M}\,1_8 \\ \mathcal{M}^* 1_8 & \sigma\cdot\vec{p}\end{pmatrix}, \qquad h_\chi(\vec{p})^2 = \big(\vec{p}^{\,2} + |\mathcal{M}|^2\big)1_{16},$$

has eigenvalues $\pm E$, $E = \sqrt{\vec{p}^{\,2} + |\mathcal{M}|^2}$, and the projector on the filled sea is $\Pi_-(\vec{p}) = \frac{1}{2}\big(1_{16} - h_\chi(\vec{p})/E\big)$, whose RL block is $-\mathcal{M}^*/(2E)\,1_8$. Per momentum $\kappa_{\vec{p}} = \operatorname{tr}\Pi_-^{RL} = -4\mathcal{M}^*/E$, and with $|S^6|/(2\pi)^7 = 1/(120\pi^4)$

$$\kappa = -4\,\mathcal{M}^* J(|\mathcal{M}|), \qquad J(M) = \int_{|\vec{p}|<\Lambda} \frac{d^7p}{(2\pi)^7}\frac{1}{\sqrt{\vec{p}^{\,2} + M^2}} = \frac{1}{120\pi^4}\int_0^\Lambda \frac{p^6\,dp}{\sqrt{p^2 + M^2}},$$
$$K(M) = \frac{1}{120\pi^4}\int_0^\Lambda p^6\sqrt{p^2 + M^2}\,dp.$$

The Hartree Kohn–Sham mass $\mathcal{M} = m + \partial e_H/\partial\kappa = m + 2g\kappa^*$ gives the gap equation and the vacuum energy density

$$\mathcal{M} = m - 8\,g\,J(|\mathcal{M}|)\,\mathcal{M},$$

$$\mathcal{E}(\mathcal{M}) = -8K(|\mathcal{M}|) + 8J(|\mathcal{M}|)\,\mathrm{Re}\big[(\mathcal{M}-m)\mathcal{M}^*\big] + 32\,g\,|\mathcal{M}|^2 J(|\mathcal{M}|)^2,$$

the sum of the sea eigenvalues $-8K$ minus the double counting $2\mathrm{Re}[(\mathcal{M}-m)\kappa]$ plus $e_H = 2g|\kappa|^2$; per momentum, $\mathrm{tr}(\Pi_- h_0) = -8E + 8\,\mathrm{Re}[(\mathcal{M}-m)\mathcal{M}^*]/E$. The stationarity condition of $\mathcal{E}$ is the gap equation; for $m = 0$ $d\mathcal{E}/d|\mathcal{M}| = 8|\mathcal{M}|\,(1+8gJ)\,(J + |\mathcal{M}|J')$. For $m = 0$ a solution $\mathcal{M} \neq 0$ exists if and only if $1 + 8gJ(|\mathcal{M}|) = 0$ has a root, i.e. if and only if

$$g < g_c = -\frac{1}{8J(0)} = -\frac{90\pi^4}{\Lambda^6}, \qquad J(0) = \frac{\Lambda^6}{720\pi^4}, \qquad K(0) = \frac{\Lambda^8}{960\pi^4}.$$

For $m = 0$, $\mathcal{E}$ depends only on $|\mathcal{M}|$: the broken vacua $\mathcal{M} = |\mathcal{M}|e^{i\phi}$ are degenerate and related by chiral rotations, the phase ϕ being the Goldstone direction. A small explicit mass $m > 0$ selects the real positive $\mathcal{M}$ ($\mathcal{E}(\mathcal{M}e^{i\phi}) - \mathcal{E}(\mathcal{M}) = 8Jm|\mathcal{M}|(1-\cos\phi) > 0$). **Example:** $\Lambda = 1$, $g = \frac{6}{5}g_c$:

quantity	**value**
$g_c = -90\pi^4$	-8766.81819306022
$g = \frac{6}{5}g_c$	-10520.1818316723
$\lvert\mathcal{M}\rvert$ for $m = 0$	0.553693851829242
$\kappa = -4\mathcal{M}J(\mathcal{M})$ ($\mathcal{M}$ real)	$-2.63157928583649 \times 10^{-5}$
$\mathcal{E}(\mathcal{M}) - \mathcal{E}(0)$	$-1.31024496290100 \times 10^{-6}$
$\mathcal{M}$ for $m = 1/100$	0.585312846746221

7+1.nb also diagonalizes $h_\chi(\vec{p})$ at one momentum, $\vec{p} = (\frac{3}{10}, -\frac{1}{5}, \frac{1}{10}, \frac{1}{4}, 0, -\frac{3}{20}, \frac{1}{20})$, with the complex mass $\mathcal{M} = |\mathcal{M}|e^{7i/10}$, and confirms $h_\chi^2 = E^2$, $\Pi_-^2 = \Pi_-$, $\mathrm{tr}\,\Pi_- = 8$, $\kappa_{\vec{p}} = -4\mathcal{M}^*/E$ and the sea energy formula.

11. Verification

11.1 In Mathematica

7+1.nb was evaluated with Wolfram Language 15.0.1 for Microsoft Windows (64-bit) (July 2, 2026), by tools/run_eval.wl (all cells evaluated in a fresh kernel, then saved), in 283.883 s, with 0 message cells. Every claim is a cell `check["name", test]`; sections 7–9 contain 74 of them and the whole notebook 479. All are True; the summary of section 10 prints `Length[checks]` = 203, `failedChecks` = {} and `allChecksPassed` = True. The checks of sections 7–9:

§	**check (as named in 7+1.nb)**	**result**
7.1	Dirac matrices: Γ17up[0] == Γ17[0], Γ17up[i] == -Γ17[i]; all REAL, every entry 0 or ±1	True
7.1	Dirac adjoint: ConjugateTranspose[Γ17up[A]] == Γ17[0].Γ17up[A].Γ17[0] for A = 0..7	True
7.1	βD == {{0,ID8},{ID8,0}} and αD[i] == {{0,τO[i]},{-τO[i],0}}: real and symmetric	True
7.1	αD[i].αD[j] + αD[j].αD[i] == 2 δij ID16, αD[i].βD + βD.αD[i] == 0, βD.βD == ID16	True
7.1	ω17 = Γ17[8] == {{0,-ID8},{ID8,0}} is real, ω17.ω17 == -ID16, and ω17 anticommutes with every Γ17[A]	True
7.2	(I Γ17up[A] ∂_A + m)(I Γ17up[A] ∂_A - m) dirac16complex == -(∂0^2 - ∂1^2 - ... - ∂7^2 + m^2) dirac16complex (16 arbitrary functions)	True
7.3	Γ17[0].(I Γ17up[A] ∂_A - m) ψ == I ∂0 ψ - h ψ with h = -I αD[i] ∂_i + βD m (16 arbitrary functions)	True
7.4	h(p) == {{0, M}, {Transpose[M], 0}} with the REAL 8×8 matrix M = m ID8 + p_i τO[i]	True
7.4	M.Transpose[M] == Transpose[M].M == E^2 ID8 and h(p).h(p) == E^2 ID16, E^2 = m^2 + p^2	True
7.4	h(p).u_s == E u_s and h(-p).v_s == -E v_s for s = 1..8	True
7.4	(pslash - m).u_s == 0 and (pslash + m).v_s == 0, pslash = E Γ17[0] + p_i Γ17[i]	True
7.4	u_s.u_r == v_s.v_r == 2E δsr, and u_s(p).v_r(-p) == 0	True
7.4	plane waves: u_s Exp[-I(E x0 - p.x)] and v_s Exp[+I(E x0 - p.x)] solve (I Γ17up[A] ∂_A - m) ψ == 0	True
7.4	example p = (2,2,4,0,0,0,0), m = 1: E = 5 and the eight u_s are real (entries in Sqrt[5] ℚ) and solve h(p).u_s == 5 u_s	True
7.5	spin sums: Σ_s u_s ubar_s == pslash + m and Σ_s v_s vbar_s == pslash - m	True
7.5	completeness: Σ_s u_s(p) u_s(p)^T + Σ_s v_s(-p) v_s(-p)^T == 2E ID16	True

§	check (as named in 7+1.nb)	result
7.5	U = (u_1..u_8, v_1(-p)..v_8(-p))/Sqrt[2E] is orthogonal and Transpose[U].h(p).U == diag(E ID8, -E ID8)	True
7.6	Grassmann: θ1 θ2 == -θ2 θ1, θ1 θ1 == 0, associativity, and ∂/∂θ2 (θ1 θ2 θ3) == -θ1 θ3	True
7.6	the three plane waves U1, U2, U3 solve the Dirac equation with m = 1	True
7.6	Grassmann field: ψ_a ψ_b + ψ_b ψ_a == 0 for all a, b (in particular ψ_a^2 == 0)	True
7.6	current conservation ∂_A (ψbar Γ17up[A] ψ) == 0 for the Grassmann-valued solution ψ = θ1 U1 + θ2 U2 + θ3 U3	True
7.7	canonical anticommutator from the mode algebra: {ψ_a(x0,x), ψ_b(x0,y)^†} == δab (1/V) Σ_p Exp[I p.(x - y)] (five momenta, symbolic x, y)	True
7.8	ψ -> ψc = ω17.ψ^*: (I Γ17up[A] ∂_A - m)(ω17.ψ^*) == ω17.((I Γ17up[A] ∂_A - m) ψ)^* for arbitrary ψ = ψre + I ψim	True
7.8	(pslash + m).ω17.u_s == 0 and ω17.u_s == Σ_r v_r (-Transpose[M]/E)_rs; ω17.ω17 == -ID16 (no Majorana spinors)	True
7.9	boost at β = 2 log 2: S == 5/4 ID16 + 3/4 Γ17[0].Γ17[1], p' = Λ.p == (55/8, -41/8, 2, 4, 0, 0, 0, 0) is on shell, (pslash' - m).S.u_s(p) == 0	True
7.10	χ17 = I ω17: Hermitian, χ17.χ17 == ID16, anticommutes with every Γ17[A], commutes with every Σ17[A,B]	True
7.10	PL17, PR17 are complex (not real); Γ17[A].PL17 == PR17.Γ17[A]; ConjugateTranspose[PL17].Γ17[0] == Γ17[0].PR17	True
7.10	EL, ER: χ17.EL == -EL, χ17.ER == ER, EL^†.EL == ER^†.ER == ID8, EL^†.ER == 0, EL.EL^† == PL17, ER.ER^† == PR17, W unitary	True
7.10	φL == (ψup + I ψdown)/Sqrt[2], φR == (I ψup + ψdown)/Sqrt[2], PL17.ψ == EL.φL, PR17.ψ == ER.φR, ψ == EL.φL + ER.φR	True
7.11	σW[i] = I τO[i]: Hermitian, σW[i].σW[j] + σW[j].σW[i] == 2 δij ID8	True
7.11	chiral (Weyl) form: W^†.h(p).W == {{-σW.p, m ID8}, {m ID8, σW.p}} (symbolic p)	True
7.11	EL^†.Γ17[0].(I Γ17up[A] ∂_A - m) ψ == (I ∂0 + τO[i] ∂_i) φL - m φR (16 arbitrary functions)	True
7.11	ER^†.Γ17[0].(I Γ17up[A] ∂_A - m) ψ == (I ∂0 - τO[i] ∂_i) φR - m φL (16 arbitrary functions)	True
7.11	Klein-Gordon for each 8-component field: (I ∂0 - τO[i] ∂_i)(I ∂0 + τO[j] ∂_j) f == -(∂0^2 - ∂1^2 - ... - ∂7^2) f (8 arbitrary functions)	True
7.11	plane-wave example p = (2,2,4,0,0,0,0), m = 1, E = 5: (E + σW.p) φL == m φR and (E - σW.p) φR == m φL for all eight u_s	True
8.1	the densities ρS, ρP, j^A, j5^A are real: Γ17[0].op is Hermitian for op = ID16, I χ17, Γ17up[A], Γ17up[A].χ17	True
8.1	18 vertex pairs; lower vertex == η17[[A+1,A+1]] upper vertex in the V and A channels	True
8.2	nonlinear field equation: left derivative ∂L_int/∂ψbar_a == -[gS (ψbar ψ) ψ + gP (ψbar Iχ17 ψ) Iχ17 ψ + gV (ψbar Γ17[A] ψ) Γ17up[A] ψ + gA (ψbar Γ17[A]χ17 ψ) Γ17up[A]χ17 ψ]_a (32 Grassmann generators)	True
8.4	h_KS(p) is Hermitian for real S, P, V_A, A_A (symbolic momentum and potentials)	True
8.4	plane wave: Γ17[0].(pslash(ε,p) - m - S - I χ17 P - Γ17up[A](V_A + χ17 A_A)) == ε ID16 - h_KS(p)	True
8.4	Hartree potentials: ∂E_H/∂φ_a^* == (v_H.φ)_a, v_H = h_KS - h with S = gS ρS, P = gP ρP, V_A = gV j_A, A_A = gA j5_A (32 variables)	True
8.5	Wick's theorem in the 560-dimensional Fock space: <Φ| :(ψbar o1 ψ)(ψbar o2 ψ): |Φ> == tr(o1 R) tr(o2 R) - tr(o1 R o2 R) for all 18 vertex pairs	True
8.6	exchange self-energy: d/dt E_x(R + t dR) at t = 0 == tr(Σ_x.dR) and E_x == tr(Σ_x.R)/2 (random complex R, dR; symbolic couplings)	True
8.6	the 256 products Γ_K are orthogonal matrices with tr(Inverse[Γ_K].Γ_L) == 16 δKL	True
8.6	Fierz completeness: Σ_K (Γ_K)_ab (Inverse[Γ_K])_cd == 16 δad δcb (as the 256×256 matrix identity Transpose[F].F == 16 ID)	True
8.6	Fierz rearrangement of every exchange term: tr(o1 R o2 R) == (1/16) Σ_K tr(Inverse[Γ_K].R) tr(o1.Γ_K.o2.R) for all 18 vertex pairs (random complex R)	True
8.7	uniform matter: tr R == ρS, tr(Γ17up[0].R) == ρ, and the densities ρP, j^i, j5^A vanish	True
8.7	the occupied plane waves give this R: (1/2) Σ_s [u_s(p) ubar_s(p) + u_s(-p) ubar_s(-p)]/(2E) == (E Γ17[0] + m ID16)/(2E)	True
8.7	uniform exchange energy: E_x(R) == -gS/32 (ρS^2 + ρ^2) + gP/32 (ρS^2 - ρ^2) - gV/32 (8ρS^2 - 6ρ^2) + gA/32 (8ρS^2 + 6ρ^2)	True
8.7	uniform exchange self-energy: Σ_x == S_x ID16 + V_x Γ17[0] with S_x == ∂E_x/∂ρS and V_x == ∂E_x/∂ρ	True
8.7	uniform Hartree-Fock == Hartree with the Fierz-transformed couplings: gS/2 ρS^2 + gV/2 ρ^2 + E_x == G_S/2 ρS^2 + G_V/2 ρ^2 (gP, gA arbitrary)	True
8.7	c7 == 8 |S^6|/(2 Pi)^7 == 1/(15 Pi^4), |S^6| == 2 Pi^(7/2)/Gamma[7/2] == 16 Pi^3/15, and ρ == c7 kF^7/7 == kF^7/(105 Pi^4)	True
8.7	closed forms: d/dkF of the integrals gives the integrands, both vanish at kF = 0, and dε_kin/dm* == ρS	True
8.7	Hartree gap equation m* == m + gS ρS(m*) solved to 45 digits; m* is a stationary point of ε(m*) at fixed kF	True
8.7	total energy == sum of KS eigenvalues - E_H (double counting) == ε_kin - (m* - m) ρS + E_H	True

§	check (as named in 7+1.nb)	result
8.7	Hugenholtz-Van Hove at saturation: E/A == ε_F == dε/dρ; E/A - m < 0 and E/A at 0.95 kF0 and 1.05 kF0 is higher	True
8.7	Hugenholtz-Van Hove away from saturation: μ == dε/dρ == ε_F at kF = 1	True
8.7	Hartree-Fock at kF = 3/4 (gP = gA = 0): G_S == -98750, G_V == 20000; m* - m == gS ρS + S_x, V_0 == gV ρ + V_x; μ == ε_F	True
9.1	chiral densities: ρS == κ + κ^*, ρP == I(κ - κ^*), j^A == jL^A + jR^A, j5^A == jR^A - jL^A (symbolic chiral orbitals)	True
9.1	the chiral blocks behind these densities: EL^†.βD.ER == ID8, EL^†.βD.EL == 0, EL^†.αD[i].EL == -σW[i], ER^†.αD[i].ER == σW[i], EL^†.αD[i].ER == 0	True
9.2	E_H (16-component form, chiral orbitals) == gS/2 (κ + κ^*)^2 - gP/2 (κ - κ^*)^2 + (gV + gA)/2 (jL.jL + jR.jR) + (gV - gA) jL.jR	True
9.3	chiral KS Hamiltonian: W^†.h_KS(p).W == {{V^L_0 - σW.(p + V^L), M}, {M^*, V^R_0 + σW.(p + V^R)}}, M = m + S + I P, V^L = V - A, V^R = V + A	True
9.3	Hartree (Wirtinger): M - m == ∂e_H/∂κ == gS ρS + I gP ρP, V^L_A == ∂e_H/∂jL^A; V^L_A == V_A - A_A and V^R_A == V_A + A_A with V_A = gV j_A, A_A = gA j5_A	True
9.3	chiral KS potentials are derivatives: ∂E_H/∂φL^* == (V^L_0 - σW[i] V^L_i) φL + (M - m) φR and ∂E_H/∂φR^* == (V^R_0 + σW[i] V^R_i) φR + (M - m)^* φL	True
9.4	e^(Iθχ17) == cosθ ID16 - sinθ ω17 is REAL and orthogonal; U.EL == e^(-Iθ) EL, U.ER == e^(Iθ) ER	True
9.4	U^†.βD.U == βD.(cos 2θ ID16 + I sin 2θ χ17); vector and axial densities invariant	True
9.4	E_H(e^(Iθχ17) ψ) - E_H(ψ) == (gS - gP)/2 [(e^(4Iθ) - 1) κ^2 + (e^(-4Iθ) - 1) κ^*2]: U(1)_A symmetric iff gP == gS	True
9.5	uniform matter in chiral form: κ == ρS/2, jL^0 == jR^0 == ρ/2, jL^i == jR^i == 0	True
9.5	(h_χ - V_0)^2 == (p^2 + m*^2) ID16: eigenvalues V_0 ± Sqrt[p^2 + m*^2], eight times each	True
9.6	|S^6|/(2 Pi)^7 == 1/(120 Pi^4); J(0) == Λ^6/(720 Pi^4); g_c == -1/(8 J(0)) == -90 Pi^4/Λ^6; J(M) -> J(0) and K(M) -> K(0) as M -> 0	True
9.6	m = 0, g = 6/5 g_c: the gap equation 1 + 8 g J(|M|) == 0 has the root |M| (32 digits); M == 2 g κ^* with κ == -4 M J; E is stationary there and lower than E(0)	True
9.6	degenerate chiral vacua for m = 0: E(|M| e^(I ϕ)) == E(|M|) for ϕ = 1, 2, 4 (the phase is the Goldstone direction)	True
9.6	explicit breaking m = 1/100: the real M == m - 8 g J(M) M solves the gap equation, E is stationary there, and every complex phase raises E	True
9.6	the sea at one momentum: h_χ^2 == E^2 ID16; Π_- is a projector of rank 8; κ_p == tr(Π_- RL block) == -4 M^*/E; tr(Π_-.h0) == -8E + 8 Re[(M - m) M^*]/E	True

11.2 Independently in Python

tools/ref_dirac16.py recomputes everything with sympy (exact arithmetic), mpmath (30 digits) and numpy: the Dirac matrices and their properties, the Dirac and Klein–Gordon identities for 16 arbitrary functions, the real spinors, spin sums, completeness and diagonalization for a symbolic momentum, the canonical anticommutators from the mode algebra, charge conjugation, the boost, the chirality theorem (rank 255), the chiral projections and the two coupled 8-component equations, Grassmann anticommutation, current conservation and the nonlinear field equation (32-generator Grassmann algebra), the Kohn–Sham Hamiltonians and their chiral form, the Hartree potentials as Wirtinger derivatives, $U(1)_A$, the Fierz completeness and rearrangement, Wick's theorem in the 560-dimensional Fock space, the uniform-matter exchange formulas, the uniform-matter numbers (Hartree, Hartree–Fock, saturation, Hugenholtz–Van Hove) and the chiral vacuum with the Dirac sea. It has 90 boolean checks, all True.

tools/verify_7p1_nb.py parses the evaluated notebook and compares: every check output True; the printed spinors $\sqrt{5}\,u_s$ against Python; the printed closed forms of section 10.1, differentiated with sympy; and 23 printed numbers against tools/ref_dirac16.py, agreeing to at least 17 significant digits:

quantity	7+1.nb (Mathematica)	ref_dirac16.py (Python)	rel. diff.
saturation k_F	0.756636150701492895	0.756636150701492895	5.4×10^{-27}
saturation ρ	$1.38810021323168775 \times 10^{-5}$	$1.38810021323168775 \times 10^{-5}$	5.6×10^{-27}
saturation m^*	0.348097340146243265	0.348097340146243265	1.8×10^{-27}
saturation ρ_S	$6.51902659853756727 \times 10^{-6}$	$6.51902659853756727 \times 10^{-6}$	7.8×10^{-28}
saturation $\mathcal{V}_0$	0.138810021323168772	0.138810021323168772	1.7×10^{-28}
saturation $E/A - m$	−0.0283214279647385759	−0.0283214279647385759	3.5×10^{-28}
saturation ε_F	0.971678572035261424	0.971678572035261424	2.5×10^{-27}
$k_F = 1$: m^*	0.0809644302776061263	0.0809644302776061263	3.1×10^{-27}

quantity	7+1.nb (Mathematica)	ref_dirac16.py (Python)	rel. diff.
$k_F = 1$: $E/A - m$	0.410864848634332171	0.410864848634332171	4.2×10^{-27}
$k_F = 1$: ε_F	1.98098486130815429	1.98098486130815429	6.2×10^{-28}
HF $k_F = \frac{3}{4}$: m^*	0.366436301981543921	0.366436301981543921	5.2×10^{-27}
HF $k_F = \frac{3}{4}$: $\mathcal{V}_0$	0.261017754731615526	0.261017754731615526	2.1×10^{-27}
HF $k_F = \frac{3}{4}$: $E/A - m$	0.0391463380161198615	0.0391463380161198615	2.9×10^{-27}
Hartree $k_F = \frac{3}{4}$: m^*	0.362973947648030198	0.362973947648030198	2.8×10^{-27}
Hartree $k_F = \frac{3}{4}$: $E/A - m$	-0.0280653758468729609	-0.0280653758468729609	3.5×10^{-27}
chiral $\kappa = \rho_S/2$	$3.25951329926878364 \times 10^{-6}$	$3.25951329926878364 \times 10^{-6}$	7.8×10^{-28}
chiral $\mathcal{M} = m^*$	0.348097340146243265	0.348097340146243265	1.8×10^{-27}
g_c	-8766.81819306021935	-8766.81819306021935	7.0×10^{-27}
g	-10520.1818316722632	-10520.1818316722632	1.1×10^{-27}
$\lvert\mathcal{M}\rvert$ $(m = 0)$	0.553693851829241691	0.553693851829241691	3.6×10^{-27}
κ $(m = 0)$	$-2.63157928583648726 \times 10^{-5}$	$-2.63157928583648726 \times 10^{-5}$	9.8×10^{-28}
$\mathcal{E}(\mathcal{M}) - \mathcal{E}(0)$	$-1.31024496290100223 \times 10^{-6}$	$-1.31024496290100223 \times 10^{-6}$	2.7×10^{-27}
$\mathcal{M}$ $(m = \frac{1}{100})$	0.585312846746220511	0.585312846746220511	6.8×10^{-27}

12. The notebook: sections 7–10 of 7+1.nb

- **7. The Dirac equation of the quantized 16-component field dirac16complex** — 7.1 The eight real Dirac matrices and the Dirac α, β; 7.2 The field dirac16complex and its Dirac equation; 7.3 Hamiltonian form; 7.4 Plane waves and the real spinors u_s, v_s; 7.5 Spin sums, completeness, and the diagonalization of h; 7.6 Grassmann algebra: anticommuting components, Lagrangian, conserved current; 7.7 Quantization: mode expansion and canonical anticommutators; 7.8 Charge conjugation; 7.9 Lorentz covariance; 7.10 Chirality: no real chirality matrix, complex chiral projections; 7.11 The two coupled 8-component equations
- **8. A 16-component Kohn–Sham theory of the self-coupled field dirac16complex** — 8.1 The self-coupled field: Lagrangian, densities, Hamiltonian; 8.2 The nonlinear field equation (Grassmann left derivatives); 8.3 Densities, sources and the Hohenberg–Kohn theorem; 8.4 The 16-component Kohn–Sham equations; 8.5 Exchange: Wick's theorem for contact couplings; 8.6 The exchange self-energy and the Fierz identities; 8.7 Example: uniform matter (no sea), Hartree and Hartree–Fock
- **9. The chiral Kohn–Sham theory (8 + 8 components)** — 9.1 Chiral densities; 9.2 The Hartree energy in chiral variables; 9.3 The chiral Kohn–Sham equations; 9.4 Chiral symmetry; 9.5 Uniform matter in chiral form; 9.6 The chiral vacuum with the Dirac sea: spontaneous chiral symmetry breaking
- **10. Summary of all checks (sections 1–9)** — the four summary cells: `Length[checks]`, `failedChecks`, `allChecksPassed` and a table of all checks

Sections 0–6 (the octonions and the Clifford algebra of 7+1 spacetime) are unchanged, except for the explanation of the signature added on 27 October 2026 at the author's request (section 2.1: some text cells, two comments and one check). The evaluated notebook as it was before sections 7–10 were appended is kept in the workspace folder backups/ (file 7+1_2026-10-26_before_dirac16complex.nb), a folder excluded from the repository by .gitignore. Section 6 summarizes the checks of sections 1–5 (it is evaluated before sections 7–9); section 10 summarizes all checks.

To re-evaluate. Open 7+1.nb in Wolfram 15.0.1, start from a fresh kernel (Evaluation ▷ Quit Kernel ▷ Local, if the kernel has been used) and choose Evaluation ▷ Evaluate Notebook (about 284 seconds), or evaluate the following line in any notebook of Wolfram 15.0.1; it evaluates 7+1.nb in place in a fresh kernel, saves it and writes build/eval_log.txt:

```
Get["D:/Developer/github/7+1/tools/run_eval.wl"]
```

The Wolfram front end used is

```
C:\Program Files\Wolfram Research\Wolfram\15.0.1\WolframNB.exe
```

The notebook itself is generated by tools/make_7p1_nb.py. The cross-check of the evaluated notebook and the independent Python checks are run with

```
python3 tools/verify_7p1_nb.py
python3 tools/ref_dirac16.py
```

Main Mathematica names. `Γ17[A]` (Γ_A), `Γ17up[A]` (Γ^A), `βD` (β), `αD[i]` (α_i), `ω17` (ω), `χ17` (χ), `PL17`, `PR17` (P_L, P_R), `EL`, `ER`, `Wch` (E_L, E_R, W), `σW[i]` (σ_i), `dirac16complex` (ψ), `DiracEq16[ψ, m]` ($(i\Gamma^A\partial_A - m)\psi$), `hD[ψ, m]` ($h\psi$), `hp[p, m]` ($h(\vec{p})$), `uS`, `vS` ($u_s$, v_s), `slash17[E, p]` ($\not{p}$), `grass`, `gMul`, `gD`, `θg[k]` (Grassmann algebra), `vPairs` (the 18 vertex pairs), `hKS[p]` ($h_{\mathrm{KS}}(\vec{p})$), `hChiKS[p]` (its chiral form), `EH16`, `EHchi` (E_H), `Exfun`, `Sigmax` (E_x, Σ_x), `products256` (Γ_K), `gap7`, `eps7`, `fermi7` (uniform matter), `Jsea`, `Ksea`, `Esea`, `gcrit`, `gNJL` (Dirac sea).

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